

FARSIGHT Software Interface Control Document (ICD) Software Design Document (SDD)

Version 1.0(DRAFT)

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1 Introduction

This document defines the interface functions and configuration parameters of the FARSIGHT system as a standalone camera module.

1.1 Purpose

This document serves as an overview of the software interface for the FARSIGHT telescope. This Interface Control Document (ICD) is not a standalone document and must be used in conjunction with the documents listed in Section 2 to ensure successful payload operations.

1.2 Scope

This document is valid with respect to the FARSIGHT system as a standalone camera system.

1.3 Usage

This document is to be used by users of the FARSIGHT payload for development of a software to interface with and operate the FARSIGHT payload as a standalone camera.

1.4 Control Authority

Upon release of version 1.0, this document is under the TURION quality control system and shall propose changes in accordance with the internal Engineering Change Order (ECO) process. Prior to version 1.0 this document is considered under a DRAFT status.

1.5 Applicable Documents

The following documents, both program and non-program, are identified within this document. Unless a version number is specified, the latest version of these documents is to be used.

1.6 Government Documentation

Future revisions of FARSIGHT's Software ICD will include modifications to ensure compliance with the Government Standards listing in Table 1: Applicable Government Documentation.

Identifier	Document Title
-	JCO, SSA, GUIDANCE, CORRECTIONS_JULY2019.pdf
-	NGA.STND.0067_1.0.MIDSI_2021-10-20.pdf
-	NATIONAL IMAGERY TRANSMISSION FORMAT 2.1: Motion Imagery Extension, version 1.3.2, 2021-10-20

Table 1: Applicable Government Documentation

1.7 Internal Documentation

Table 2 lists TURION documents and authorized changes form a part of this document.

Identifier	Document Title
CM-00945	EXPORT CONTROL, GUIDELINES, DOCUMENT MARKING
CM-01980	ICD, ELECTRICAL, FARSIGHT

Table 2: Applicable Internal Documentation

1.8 Farsight Payload System Overview

FARSIGHT is an optical payload being designed for mid- to long-range non-Earth Imaging (NEI) to support 5-year missions in Low-Earth Orbit (LEO) and Geostationary Orbit (GEO). Design, fabrication and testing of FARSIGHT is funded under a USG SBIR award. Within the context of this document and based on the current software interface design, FARSIGHT represents a standalone camera.

1.9 Mission and Company Profile

TURION was founded to accelerate the commercialization of space while focusing on key technological risks toward advancement of the greater vision of a prosperous and stable space economy. To address these risks, TURION uses a hardware-first approach to rapidly develop and iterate on space vehicles using a capability-based architecture to reduce risk and iteration cycle time.

1.10 FARSIGHT Compliance

FARSIGHT must adhere to the guidelines and requirements specified in this document and the DROID-Alpha Hosted Payload User Guide. This ensures successful integration and operation within the DROID-Alpha Satellite Bus Payload environment.

2 Farsight Interface Control Document (ICD)

The Farsight Interface Control Document (ICD) provides comprehensive technical specifications for the Farsight payload system. This document is organized into three main sections:

- **Section 3 – Farsight Firmware Architecture:** Describes the embedded firmware design, including operational modes, bootloader architecture, TMTC protocol specifications, command interface, and register definitions.
- **Section 4 – Farsight Camera Sensor Payload:** Details the Sony IMX531 camera sensor specifications, imaging capabilities, and sensor configuration parameters.
- **Section 5 – Farsight Hardware Architecture:** Covers the physical interfaces, electrical connections, and hardware block diagrams of the Farsight payload system.

3 Farsight Firmware Architecture

Farsight Firmware is designed such that the entire payload system act as a standalone camera module. The following sections describe the architecture of the Farsight Firmware.

3.1 Firmware Modes of operation

Figure 1 shows the state diagram for the Farsight firmware operation modes.

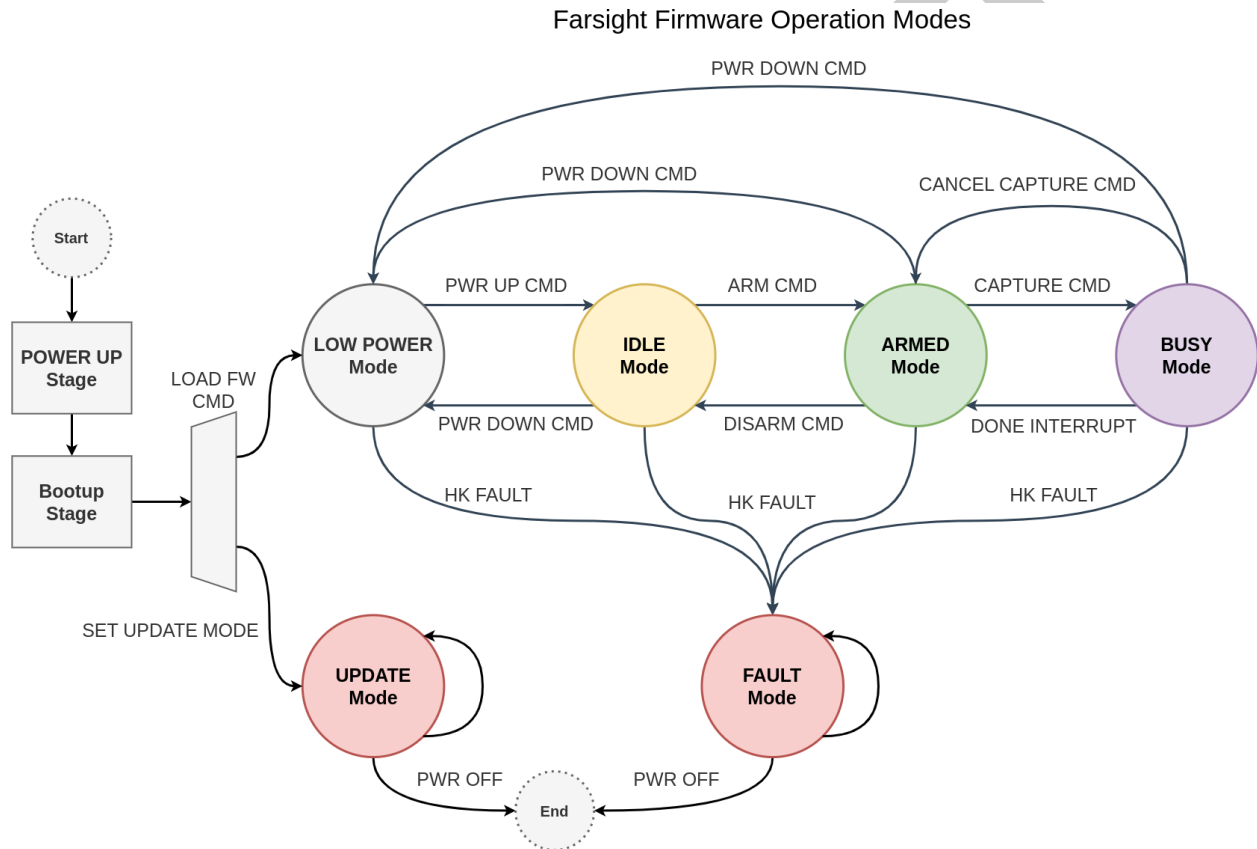


Figure 1: Farsight Firmware Operation Modes

The Farsight firmware operates as a finite state machine with the following distinct operational modes:

- **Low Power (CAMERA_LP_STATE):** The default mode after a reset and successful boot of critical power regions. FARSIGHT can also be commanded to enter this mode. Within this mode, the camera sensor is not powered, health telemetry acquisition begins, and FARSIGHT awaits a command for mode transition. From Low Power State, FARSIGHT can be commanded to enter Idle State.
- **Idle (CAMERA_IDLE_STATE):** Upon request, FARSIGHT will transition from Low Power State to Idle State, where FARSIGHT will prepare for image acquisition commands and nominal payload operations. When FARSIGHT enters this state, the CMOS sensor and all required interfaces for imaging operations will be enabled to prepare for imaging commands. From Idle State, FARSIGHT can transition to Armed State or return to Low Power State.
- **Armed (CAMERA_ARMED_STATE):** A preparatory state where the system is configured and awaiting a capture trigger. The imaging subsystem is primed and ready for immediate operation. In this state, FARSIGHT awaits either a user-initiated cue or a scheduled capture event to begin image acquisition.

From Armed State, FARSIGHT can transition to Busy State upon receiving a capture command, or return to Idle State via the disarm command.

- **Busy (CAMERA_BUSY_STATE):** Upon request, FARSIGHT will transition from Armed State to Busy State, where the system actively captures images and stores them into the designated DDR4 frame buffer space. Further details about the imaging operation are provided in the subsequent SDD. During imaging, certain camera sensor parameters can be updated via SPI commands; some updates will be applied immediately to the current frame, while others will take effect on the next frame. The specifics of parameter update timing are described in later sections. Upon completion of image capture, the FPGA hardware generates an interrupt to the RISC-V firmware indicating capture completion, at which point FARSIGHT transitions from Busy State back to Armed State.
- **Fault (CAMERA_FAULT_STATE):** A safety state entered when a housekeeping (HK) fault is detected. This state can be reached from Low Power, Idle, or Armed states when the system detects anomalous conditions that require immediate attention.

The firmware follows a staged power-up sequence (Low Power → Idle → Armed → Busy), allowing controlled resource allocation and power management. Direct power-down capability from any active state ensures rapid response to system shutdown requests.

3.2 Firmware TMTC Specification

In this section we will define the Telemetry and Telecommand (TMTC) interface between the host computer (HOST) and Farsight Avionics computer (FAV). Before going over the details let's go over some terminologies used in this section.

Table 3: Software TMTC Interface Terminologies

Term	Description
Host	Host is the computer that communicates with the Farsight Avionics Computer by sending telecommands (TC) and receiving telemetry (TL).
TMTC Packet	Telemetry and Telecommand packet is a formatted unit of data that is transmitted between the Host and FAV. TMTC Packet consists of a Header and a Datagram.
Datagram	Datagram or payload is the actual information or message being sent between FAV and Host.
Command	Commands are a set of instructions sent from the host to the Farsight Avionics computer to control operations and configuration of the FARSIGHT payload.
Science Payload	FARSIGHT science data payload are images captured by the FARSIGHT camera sensor.

TMTC Serial interface setting information is provided below:

- **Stop Bits:** 1
- **Parity:** None
- **Data Size:** 8 bits
- **Baud Rate:** 115200 (default, configurable)

FARSIGHT has a custom packetization protocol whose packet format is defined in Figure 2. This protocol operates in a synchronous point-to-point request-reply pattern in which the host (computer interfacing with FARSIGHT) will send a command and then receive a response from Farsight Avionics Computer (FAV) based on that command. This packet information is always sent from the most significant byte (MSB) first.

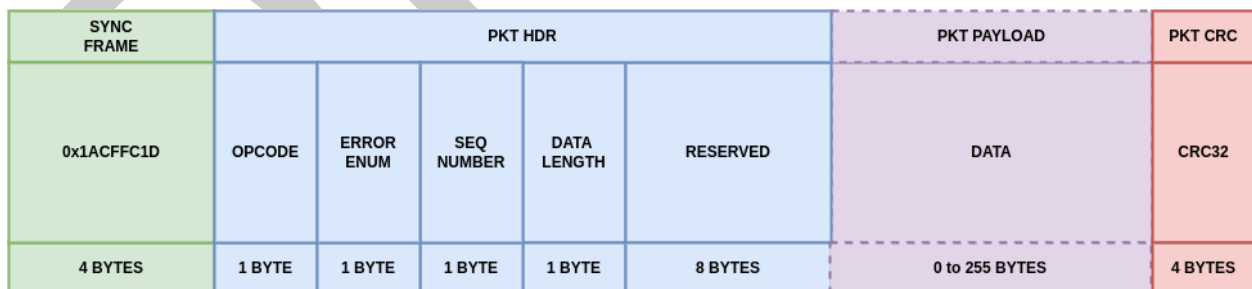


Figure 2: Farsight Avionics TMTC Packet Format

Table 4 describes the TMTC packet structure in detail.

The CRC-32 is calculated over all bytes except sync frame for data integrity verification.

Table 4: FARSIGHT TMTC Packet Structure

Field	Size (bytes)	Description
Sync Frame	4	Fixed synchronization pattern: 0x1ACFFC1D (transmitted as 0x1A, 0xCF, 0xFC, 0x1D)
Opcode	1	Command or function code identifying the operation
Error Code	1	Error status (used in reply packets from FAV)
Sequence Number	1	Packet sequence counter for tracking
Data Length	1	Length of the data payload in bytes
Reserved	8	Reserved for future use
Data Payload	Variable	Command-specific data (0 to 255 bytes)
CRC32	4	32-bit CRC calculated over all bytes except the sync frame

3.2.1 TMTC Byte Order

The FARSIGHT TMTC protocol uses mixed byte ordering. All packet framing and header fields (sync frame, reserved, and CRC) are transmitted in **big-endian** (most significant byte first) order. Single-byte fields (opcode, error code, sequence number, and data length) have no byte-order dependency. The data payload is transmitted in **little-endian** (least significant byte first) order, consistent with the native byte order of the RISC-V processor on the Farsight Avionics Computer.

Table 5 summarizes the byte order for each packet field.

Table 5: TMTC Packet Field Byte Order

Field	Size (bytes)	Byte Order
Sync Frame	4	Big-endian
Opcode	1	—
Error Code	1	—
Sequence Number	1	—
Data Length	1	—
Reserved	8	Big-endian
Data Payload	Variable	Little-endian
CRC32	4	Big-endian

Example. The following example illustrates the wire-level byte layout for a command packet carrying a 4-byte data payload with the value 10000 (0x00002710). Note how the data payload bytes appear in little-endian order (0x10, 0x27, 0x00, 0x00), while the sync frame and CRC remain in big-endian order.

Table 6: Wire-Level Byte Order Example (Data Value = 10000)

Field	Byte Order	Bytes on Wire
Sync Frame	Big-endian	0x1A, 0xCF, 0xFC, 0x1D
Opcode	—	0x03
Error Code	—	0x00
Sequence Number	—	0x00
Data Length	—	0x04
Reserved	—	8 bytes of 0x00
Data Payload	Little-endian	0x10, 0x27, 0x00, 0x00
CRC32	Big-endian	4 bytes (computed)

3.3 Farsight Firmware Commands

FARSIGHT's commands are defined within this section. Many of these commands will lead to specific operations guided by the register map defined in FARSIGHT REGISTERS.

THIS SECTION IS WORK IN PROGRESS

Command Name	Opcode	Short Description	Details
RESET_SYS_CMD	0x00	Resets the FARSIGHT subsystems	Link
REG_READ_SYS_CMD	0x01	Reads a specified register value	Link
REG_WRITE_SYS_CMD	0x02	Writes a value to a specified register	Link
SET_TIME_SYS_CMD	0x03	Sets FARSIGHT's absolute system time	Link
START_ACQ_SYS_CMD	0x04	Starts an image acquisition	Link
FOCUS_SYS_CMD	0x05	Initiates a focus operation	Link
XFER_BUFF_SYS_CMD	0x06	Transfers a section of the DDR4 frame buffer	Link
UPDATE_CALIB_SYS_CMD	0x07	Update focus mechanism calibration table	Link
UPDATE_FW_SYS_CMD	0x08	Update Farsight Firmware	Link
ECHO_SYS_CMD	0xFF	Echoes received payload for test and validation	Link

Table 7: FARSIGHT Command Interface Summary

3.3.1 Firmware Command Descriptions

Cmd Name	RESET_SYS_CMD
Description	This command will perform a reset of various FARSIGHT subsystems
Opcode	0x00
Arguments	Argument 1: Name: <code>target_reset</code> Type: <code>uint8_t</code> Length (bytes): 1 Description: Bit field to specify the target subsystem to be reset. • Bit 0: RESET_CAM_SENSOR, setting this bit will put the Sensor in the Low Power mode (Camera will be turned off). • Bit 1: RESET_FOUCS_MECH, setting this bit will reset all the focus mechanism faults and stop driving the motor controllers at their current position. • Bit 2: RESET_REG_SPACE, setting this bit will reset all the firmware registers to their default values. • Bit 3: RESET_MEM_BUFF, setting this bit will reset the image frame memory buffer to index 0. • Bit 7:4: Not used. Valid value: 0x00 to 0x0F
Return Value	None

Table 8: Command: RESET_SYS_CMD

Cmd Name	REG_READ_SYS_CMD
Description	Reads a register value.
Opcode	0x01
Arguments	Argument 1: Name: reg_id Type: uint8_t Length (bytes): 1 Description: Register ID to read. Valid value: 0x00 to 0xFF
Return Value	uint32_t

Table 9: Command: REG_READ_SYS_CMD

Cmd Name	REG_WRITE_SYS_CMD
Description	Writes a register value.
Opcode	0x02
Arguments	Argument 1: Name: reg_id Type: uint8_t Length (bytes): 1 Description: Register ID to write. Valid value: 0x00 to 0xFF (For addresses not mapping to any register and error will be returned) Argument 2: Name: reg_value Type: uint32_t Length (bytes): 4 Description: Register value. Valid value: Varied based on the register's allowable values.
Return Value	None

Table 10: Command: REG_WRITE_SYS_CMD

Cmd Name	SET_TIME_SYS_CMD
Description	After receiving this command the time stamp of the FARSIGHT will be set on the next rising edge of the PPS signal.
Opcode	0x03
Arguments	Argument 1: Name: <code>TIMESTAMP_UNIX_EPOCH_SEC</code> Type: <code>uint32_t</code> Length (bytes): 4 Description: Whole seconds of the UTC timestamp expressed as the number of seconds elapsed since the UNIX epoch (January 1, 1970 00:00:00 UTC). This value is latched on the next rising edge of the PPS signal. Valid value: <code>0x0000_0000</code> to <code>0xFFFF_FFFF</code> Argument 2: Name: <code>TIMESTAMP_SUBSEC_USEC</code> Type: <code>uint32_t</code> Length (bytes): 4 Description: Sub-second portion of the UTC timestamp in microseconds. Combined with <code>TIMESTAMP_UNIX_EPOCH_SEC</code>, this provides microsecond-resolution time synchronization. This value is latched on the next rising edge of the PPS signal. Valid value: <code>0x0000_0000</code> to <code>0x000F_4240</code> (0 to 999 999)
Return Value	None

Table 11: Command: SET_TIME_SYS_CMD

Cmd Name	START_ACQ_SYS_CMD
Description	Requests Farsight to take pictures upon receiving this command. This command will only take effect if register <code>SENSOR_TRIG_MODE_SYS_REG</code> is set to <code>TRIG_ON_CMD</code> value. For other configurations and error will be returned
Opcode	0x04
Arguments	None
Return Value	None

Table 12: Command: START_ACQ_SYS_CMD

Cmd Name	FOCUS_SYS_CMD
Description	Calculates a focus distance and commands motion to that position.
Opcode	0x05
Arguments	Argument 1: Name: <code>raw_pos</code> Type: <code>uint32_t</code> Length (bytes): 4 Description: Focus calculation input. Valid value: TBD
Return Value	TBD

Table 13: Command: FOCUS_SYS_CMD

Cmd Name	XFER_BUFF_SYS_CMD
Description	TBD
Opcode	0x06
Arguments	Argument 1: Name: image_index Type: uint16_t Length (bytes): 2 Description: Image index. Range from 0 to 767 where the first 0 to 511 indexes will point to DDR4 16GB and 512 to 767 indexes will refer to DDR4 8GB. Valid value: 0 to 767 Argument 2: Name: num_images Type: uint16_t Length (bytes): 2 Description: Number of images to transfer. If the requested number of images is larger than max memory buffer an error will be returned. If index + requested number of images is greater than max index range an error will be returned. Valid value: 1 to 768
Return Value	None

Table 14: Command: XFER_BUFF_SYS_CMD

Cmd Name	UPDATE_CALIB_SYS_CMD
Description	Update FARSIGHT focus mechanism calibration table.
Opcode	0x07
Arguments	TBD
Return Value	TBD

Table 15: Command: UPDATE_CALIB_SYS_CMD

Cmd Name	UPDATE_FW_SYS_CMD
Description	Update Farsight firmware. TBD
Opcode	0x08
Arguments	TBD
Return Value	TBD

Table 16: Command: UPDATE_FW_SYS_CMD

Cmd Name	ECHO_SYS_CMD
Description	Reflects the received packet data for testing.
Opcode	0xFF
Arguments	Argument 1: Name: payload Type: uint8_t Length (bytes): Variable Description: Data echoed in the response. Valid value: N/A

Table 17: Command: ECHO_SYS_CMD

3.4 Farsight Firmware Error Enumeration

TODO: This section is to be completed.

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3.5 Firmware Registers

Table 18 provides a summary of all FARSIGHT firmware registers. Each register is assigned a unique identifier that should be used when requesting access to the register via register read and write commands. Click the *Description* link to navigate to the detailed register description.

Register Name	Reg ID	Access	Short Description	Details
SENSOR_STATE_SYS_REG	0x00	RO	Camera sensor operation status	Link
SENSOR_PXL_READOUT_FORMAT_SYS_REG	0x04	RW	Sensor frame output format	Link
SENSOR_READ_DIR_SYS_REG	0x08	RW	Sensor readout direction	Link
SENSOR_BIT_DEPTH_SYS_REG	0x0C	RO	Sensor image bit-depth	Link
SENSOR_GAIN_SYS_REG	0x10	RW	Sensor gain in dB ($\times 10$)	Link
SENSOR_BLO_SYS_REG	0x14	RW	Black level offset value	Link
SENSOR_EXPO_USEC_SYS_REG	0x18	RW	Exposure time in microseconds	Link
SENSOR_NUM_FRM_TO_CAPTURE_SYS_REG	0x1C	RW	Frames to capture per trigger	Link
SENSOR_TRIG_MODE_SYS_REG	0x20	RW	Image trigger mode configuration	Link
SENSOR_TEMP_RAW_SYS_REG	0x24	RO	Raw sensor temperature reading	Link
SENSOR_OVER_TEMP_RAW_HYST_SYS_REG	0x28	RW	Over-temperature hysteresis threshold	Link
BUFF_WRITE_INDEX_SYS_REG	0x2C	RW	DDR4 frame buffer write pointer index	Link
FOCUS_REQ_DIST_M_SYS_REG	0x30	RO	Requested focus distance	Link
FOCUS_COMP_DIST_M_SYS_REG	0x34	RO	Computed focus distance from LVDT	Link
FOCUS_LVDT_POS_NM_REG	0x38	RO	LVDT sensor position in nanometers	Link
HOUSEKEEPER_STATUS_SYS_REG	0x3C	RO	Power region status bitfield	Link
TIMESTAMP_UNIX_EPOCH_SEC_SYS_REG	0x40	RO	Unix epoch timestamp in seconds	Link
TIMESTAMP_SUBSEC_NSEC_SYS_REG	0x44	RO	Sub-second timestamp in nanoseconds	Link
ETH1_SRC_MAC_ADDR_HIGH_SYS_REG	0x48	RW	ETH1 source MAC address [47:32]	Link
ETH1_SRC_MAC_ADDR_LOW_SYS_REG	0x4C	RW	ETH1 source MAC address [31:0]	Link
ETH1_DEST_MAC_ADDR_HIGH_SYS_REG	0x50	RW	ETH1 destination MAC address [47:32]	Link
ETH1_DEST_MAC_ADDR_LOW_SYS_REG	0x54	RW	ETH1 destination MAC address [31:0]	Link
ETH1_SRC_IP_ADDR_SYS_REG	0x58	RW	ETH1 source IPv4 address	Link
ETH1_DEST_IP_ADDR_SYS_REG	0x5C	RW	ETH1 destination IPv4 address	Link
ETH1_SRC_UDP_PORT_SYS_REG	0x60	RW	ETH1 source UDP port	Link
ETH1_DEST_UDP_PORT_SYS_REG	0x64	RW	ETH1 destination UDP port	Link
NOOP_SYS_REG	0xFE	RW	No Operation register for testing	Link
VERSION_SYS_REG	0xFF	RO	Subsystem version info	Link

Table 18: Firmware Register Summary

3.5.1 Firmware Registers Description

This section includes tabulated descriptions of Farsight user registers.

Reg Name	SENSOR_STATE_SYS_REG
Description	The current operation status of FARSIGHT's camera sensor.
ID	0x00
Access	Read Only
Data Type	ENUM
ENUM Desc.	• 0: LP_MODE: Sensor is not powered on. • 1: IDLE_MODE: Sensor is powered on and configurable, but not ready to take images. • 2: ARMED_MODE: Sensor is capable of capturing images and is waiting for trigger. • 3: BUSY_MODE: Sensor is currently capturing images. • 4: FAULT_MODE: Firmware has encountered a fault and entered a protective state.
Reset Value	0: LP_MODE
Min Value	0
Max Value	4

Table 19: Register: SENSOR_STATE_SYS_REG

Reg Name	SENSOR_PXL_READOUT_FORMAT_SYS_REG
Description	Sensor frame output format.
ID	0x04
Access	Read/Write
Data Type	ENUM
ENUM Desc.	• 0: ALL_PIXEL: All pixels included. Image Size 4504x4504 pixel. • 1: 2x2_BINNING: 2 by 2 binning format. Image size 2252 x 2252 pixels. • 2: HALF_SAMPLING: Half sampling format. Image size 2252 x 2252 pixels.
Reset Value	0: ALL_PIXEL
Min Value	0
Max Value	2

Table 20: Register: SENSOR_PXL_READOUT_FORMAT_SYS_REG

Reg Name	SENSOR_READ_DIR_SYS_REG
Description	Sensor vertical and horizontal readout direction
ID	0x08
Access	Read/Write
Data Type	ENUM
ENUM Desc.	• 0: READ_DIR_HORIZ_NORM_VERT_NORM: Normal horizontal and vertical readout. • 1: READ_DIR_HORIZ_NORM_VERT_INVR: Normal horizontal and inverted vertical readout. • 2: READ_DIR_HORIZ_INVR_VERT_NORM: Inverted horizontal and normal vertical readout. • 3: READ_DIR_HORIZ_INVR_VERT_INVR: Inverted horizontal and vertical readout.
Reset Value	0: READ_DIR_HORIZ_NORM_VERT_NORM
Min Value	0
Max Value	3

Table 21: Register: SENSOR_READ_DIR_SYS_REG

Reg Name	SENSOR_BIT_DEPTH_SYS_REG
Description	The bit-depth of the images that FARSIGHT's sensor will output. <i>The bit-depth parameter has been identified as a future improvement for FARSIGHT. Until further notice, this bit-depth will be locked at 12-bits.</i>
ID	0x0C
Access	Read Only
Data Type	ENUM
ENUM Desc.	• 0: PIXEL_BIT_DEPTH_12: Width of each pixel will be 12 bits.
Reset Value	0: PIXEL_BIT_DEPTH_12
Min Value	0
Max Value	2

Table 22: Register: SENSOR_BIT_DEPTH_SYS_REG

Reg Name	SENSOR_GAIN_SYS_REG
Description	Gain in dB represented by an integer 10x the desired gain. When the user sets this value, the sensor will divide the value by 10. Example: User sets SENSOR_GAIN_SYS_REG to 345, resulting gain is 34.5 dB.
ID	0x10
Access	Read/Write
Data Type	uint32_t
Reset Value	0
Min Value	0
Max Value	480

Table 23: Register: SENSOR_GAIN_SYS_REG

Reg Name	SENSOR_BLO_SYS_REG
Description	Sensor Black level offset value.
ID	0x14
Access	Read/Write
Data Type	uint32_t
Reset Value	240 (This value is recommended for 12-bit operation)
Min Value	0
Max Value	4095

Table 24: Register: SENSOR_BLO_SYS_REG

Reg Name	SENSOR_EXPO_USEC_SYS_REG
Description	Sensor exposure in microseconds.
ID	0x18
Access	Read/Write
Data Type	uint32_t
Reset Value	500
Min Value	0 represents minimum exposure time of the sensor (50 nanoseconds)
Max Value	$2^{22} = 4,194,304$ uSec

Table 25: Register: SENSOR_EXPO_USEC_SYS_REG

Reg Name	SENSOR_NUM_FRM_TO_CAPTURE_SYS_REG
Description	Total number of frames to capture per trigger start request. Setting this value to 0 will indicate continuous capturing and filling up the circular memory buffer.
ID	0x1C
Access	Read/Write
Data Type	uint32_t
Reset Value	0
Min Value	0
Max Value	$2^{10} = 1024$

Table 26: Register: SENSOR_NUM_FRM_TO_CAPTURE_SYS_REG

Reg Name	SENSOR_TRIG_MODE_SYS_REG
Description	Configuration of FARSIGHT's image trigger mode. Based on this value, FARSIGHT will watch for the respective trigger event prior to initiating image capture.
ID	0x20
Access	Read/Write
Data Type	ENUM
ENUM Desc.	0: TRIG_ON_CMD: Start exposure immediately after receiving a START_ACQ_SYS_CMD command. 1: TRIG_SCHED: Schedule a trigger at a given time. FARSIGHT will wait until the requested time is reached. Time accuracy depends on the integrity of the PPS signal and could vary up to $\pm 100 \mu s$ (To be verified) 2: TRIG_EXTERNAL: Start exposure immediately after a configured GPIO's rising edge has been detected.
Reset Value	0: TRIG_ON_CMD
Min Value	0
Max Value	2

Table 27: Register: SENSOR_TRIG_MODE_SYS_REG

Reg Name	SENSOR_TEMP_RAW_SYS_REG
Description	Raw 8-bit reading of the image sensor's thermometer. The way to calculate the temperature is shown below. $\text{Temperature } [^{\circ}\text{C}] = \frac{\text{TEMP_VAL} - 51.784}{1.3125}$ Compatible temperature after conversion to degrees Celsius: $T_f = -30^{\circ}\text{C to } 105^{\circ}\text{C} \quad (\text{Accuracy: } \pm 5^{\circ}\text{C})$
ID	0x24
Access	Read Only
Data Type	uint32_t
Reset Value	0
Min Value	0
Max Value	255

Table 28: Register: SENSOR_TEMP_RAW_SYS_REG

Reg Name	SENSOR_OVER_TEMP_RAW_HYST_SYS_REG
Description	Raw 16-bit ADC reading marking an over-temperature hysteresis event. If this condition persists for 1 second (TBR), then a transition to LP_MODE will automatically be triggered. Setting this value to 255 will disable this feature. The provided target temperature value must be converted to the RAW format before sent to FARSIGHT using the following formula: $\text{TEMP_VAL} = \min(255, \max(0, \lfloor 1.3125 \cdot \text{Temperature} + 51.784 \rfloor))$
ID	0x28
Access	Read/Write
Data Type	uint32_t
Reset Value	0
Min Value	0
Max Value	255

Table 29: Register: SENSOR_OVER_TEMP_RAW_HYST_SYS_REG

Reg Name	BUFF_WRITE_INDEX_SYS_REG
Description	Current index of the write pointer used by the FARSIGHT FPGA to write data into the DDR4 frame buffer. The FARSIGHT avionics system contains two separate DDR4 memory spaces, with a total capacity of 24 GB, organized as follows: • Index 0 to 511: 16 GB DDR4 memory space • Index 512 to 767: 8 GB DDR4 memory space
ID	0x2C
Access	Read/Write
Data Type	uint32_t
Reset Value	0
Min Value	0
Max Value	767

Table 30: Register: BUFF_WRITE_INDEX_SYS_REG

Reg Name	FOCUS_REQ_DIST_M_SYS_REG
Description	Focus distance requested by the user via a FOCUS_SYS_CMD. This is the only mechanism by which this register can be updated.
ID	0x30
Access	Read Only
Data Type	uint32_t
Reset Value	N/A
Min Value	5000
Max Value	10000000

Table 31: Register: FOCUS_REQ_DIST_M_SYS_REG

Reg Name	FOCUS_COMP_DIST_M_SYS_REG
Description	Computed focus distance derived from the LVDT position using a calibrated lookup table (LUT).
ID	0x34
Access	Read Only
Data Type	uint32_t
Reset Value	Computed on startup after the first valid LVDT read
Min Value	5000
Max Value	10000000

Table 32: Register: FOCUS_COMP_DIST_M_SYS_REG

Reg Name	FOCUS_LVDT_POS_NM_REG
Description	Measured LVDT sensor position expressed in nanometers. This value is used to derive the computed focus distance.
ID	0x38
Access	Read Only
Data Type	int32_t
Reset Value	Read on startup
Min Value	TBD
Max Value	TBD

Table 33: Register: FOCUS_LVDT_POS_NM_REG

Reg Name	HOUSEKEEPER_STATUS_SYS_REG
Description	Power Regions status per housekeeper monitor report. The status is in Bitfield format [11:0]. Details of bit interpretation is described below. Note 1: All status bits are active high meaning 1 means powered on. Note 2: All other bits not specified below are tied to 0.
ID	0x3C
Access	Read Only
Data Type	uint32_t
Bitfield Desc.	• Bit 0: ProASIC3 FPGA power status. • Bit 1: Stepdown power-converter power status. • Bit 2: 8GB DDR4 power status. • Bit 3: 16GB DDR4 power status. • Bit 4: PolarFire FPGA power status. • Bit 5: LVDS power status. • Bit 6: ETH1 PHY power status. • Bit 7: ETH2 PHY power status. • Bit 8: Primary Stepper Motor power status. • Bit 9: Secondary Stepper Motor power status. • Bit 10: LVDT power status. • Bit 11: IMX Camera Sensor power status.
Reset Value	0x0000_0000
Min Value	0x0000_0000
Max Value	0x0000_03FF

Table 34: Register: HOUSEKEEPER_STATUS_SYS_REG

Reg Name	TIMESTAMP_UNIX_EPOCH_SEC_SYS_REG
Description	FARSIGHT's timestamp using the Unix epoch in sec since January 1, 1970, at 00:00:00 UTC. After receiving the SET_TIME_SYS_CMD the value will be set on the next rising edge of the PPS (pulse per second signal).
ID	0x40
Access	Read Only
Data Type	uint32_t
Reset Value	0x0000_0000
Min Value	0x0000_0000
Max Value	0xFFFF_FFFF

Table 35: Register: TIMESTAMP_UNIX_EPOCH_SEC_SYS_REG

Reg Name	TIMESTAMP_SUBSEC_NSEC_SYS_REG
Description	Sub-second component of FARSIGHT's timestamp. This register reports the time-of-second in nanoseconds corresponding to the seconds register. The value resets to 0 when seconds increments (internal PPS boundary). The timing accuracy is managed by the FPGA system, which derives the sub-second value from the 50 MHz system clock and continuously adjusts it to maintain alignment with the PPS (pulse per second signal).
ID	0x44
Access	Read Only
Data Type	uint32_t
Reset Value	0
Min Value	0
Max Value	999,999,999

Table 36: Register: TIMESTAMP_SUBSEC_NSEC_SYS_REG

Reg Name	ETH1_SRC_MAC_ADDR_HIGH_SYS_REG
Description	Bits 47:32 of FARSIGHT's MAC address for Ethernet 1.
ID	0x48
Access	Read/Write
Data Type	uint32_t
Reset Value	0x0000_FAA4 (FA:A4)
Min Value	0x0000_0000
Max Value	0xFFFF_FFFF

Table 37: Register: ETH1_SRC_MAC_ADDR_HIGH_SYS_REG

Reg Name	ETH1_SRC_MAC_ADDR_LOW_SYS_REG
Description	Bits 31:0 of FARSIGHT's MAC address for Ethernet 1.
ID	0x4C
Access	Read/Write
Data Type	uint32_t
Reset Value	0x5196_77FA (51:96:77:FA)
Min Value	0x0000_0000
Max Value	0xFFFF_FFFF

Table 38: Register: ETH1_SRC_MAC_ADDR_LOW_SYS_REG

Reg Name	ETH1_DEST_MAC_ADDR_HIGH_SYS_REG
Description	Bits 47:32 of MAC address of the device connected to FARSIGHT Ethernet 1.
ID	0x50
Access	Read/Write
Data Type	uint32_t
Reset Value	0x0000_FAA4 (FA:A4)
Min Value	0x0000_0000
Max Value	0xFFFF_FFFF

Table 39: Register: ETH1_DEST_MAC_ADDR_HIGH_SYS_REG

Reg Name	ETH1_DEST_MAC_ADDR_LOW_SYS_REG
Description	Bits 31:0 of MAC address of the device connected to FARSIGHT Ethernet 1.
ID	0x54
Access	Read/Write
Data Type	uint32_t
Reset Value	0x5196_77FB (51:96:77:FB)
Min Value	0x0000_0000
Max Value	0xFFFF_FFFF

Table 40: Register: ETH1_DEST_MAC_ADDR_LOW_SYS_REG

Reg Name	ETH1_SRC_IP_ADDR_SYS_REG
Description	FARSIGHT's IPv4 address for Ethernet 1.
ID	0x58
Access	Read/Write
Data Type	uint32_t
Reset Value	0xC0A8_010E (192.168.1.14)
Min Value	0x0000_0000
Max Value	0xFFFF_FFFF

Table 41: Register: ETH1_SRC_IP_ADDR_SYS_REG

Reg Name	ETH1_DEST_IP_ADDR_SYS_REG
Description	IPv4 address of destination connected to FARSIGHT Ethernet 1.
ID	0x5C
Access	Read/Write
Data Type	uint32_t
Reset Value	0xC0A8_010F (192.168.1.15)
Min Value	0x0000_0000
Max Value	0xFFFF_FFFF

Table 42: Register: ETH1_DEST_IP_ADDR_SYS_REG

Reg Name	ETH1_SRC_UDP_PORT_SYS_REG
Description	FARSIGHT's UDP port for Ethernet 1.
ID	0x60
Access	Read/Write
Data Type	uint32_t
Reset Value	49152
Min Value	0
Max Value	65535

Table 43: Register: ETH1_SRC_UDP_PORT_SYS_REG

Reg Name	ETH1_DEST_UDP_PORT_SYS_REG
Description	UDP port of destination connected to FARSIGHT Ethernet 1.
ID	0x64
Access	Read/Write
Data Type	uint32_t
Reset Value	12000
Min Value	0
Max Value	65535

Table 44: Register: ETH1_DEST_UDP_PORT_SYS_REG

Reg Name	NOOP_SYS_REG
Description	General-purpose read/write register with no hardware side effects. Used for verifying TMTC register read/write functionality.
ID	0xFE
Access	Read/Write
Data Type	uint32_t
Reset Value	0x0000_0000
Min Value	0x0000_0000
Max Value	0xFFFF_FFFF

Table 45: Register: NOOP_SYS_REG

Reg Name	VERSION_SYS_REG
Description	Farsight subsystem's version.
ID	0xFF
Access	Read Only
Data Type	uint32_t
Bitfield Desc.	• Bit 7:0: RISC-V Firmware Version • Bit 15:8: PolarFire FPGA Version. • Bit 23:16: ProASIC3 FPGA Version. • Bit 31:24: Farsight deployed Avionics Board ID.
Reset Value	0
Min Value	0x0000_0000
Max Value	0xFFFF_FFFF

Table 46: Register: VERSION_SYS_REG

3.6 Farsight Firmware Image Metadata

Each image frame captured by FARSIGHT is prepended with a fixed-size metadata block. The metadata records the sensor and system configuration at the time of capture, to enable post-processing tools to fully interpret the accompanying image data without requiring out-of-band information.

All metadata fields are 32-bit wide. For fields that correspond to firmware registers, the data type and value interpretation are identical to the register description provided in Section 18. Refer to the linked register table for details on enumeration values, bitfield layouts, and valid ranges.

Table 47 describes the metadata format.

Field Name	Byte Offset	Short Description	Details
METADATA_START_FLAG	0x00	Start flag, fixed 0x4D455441 ("META")	–
METADATA_SIZE_BYTES	0x04	Total metadata block size in bytes	–
SENSOR_PXL_READOUT_FORMAT	0x08	Sensor frame output format	Link
SENSOR_READ_DIR	0x0C	Sensor readout direction	Link
SENSOR_BIT_DEPTH	0x10	Sensor image bit-depth	Link
SENSOR_GAIN	0x14	Sensor gain in dB ($\times 10$)	Link
SENSOR_BLO	0x18	Black level offset value	Link
SENSOR_EXPO_USEC	0x1C	Exposure time in microseconds	Link
SENSOR_TRIG_MODE	0x20	Image trigger mode configuration	Link
SENSOR_TEMP_RAW	0x24	Raw sensor temperature reading	Link
BUFF_WRITE_INDEX	0x28	DDR4 frame buffer write pointer index	Link
FOCUS_REQ_DIST_M	0x2C	Requested focus distance	Link
FOCUS_COMP_DIST_M	0x30	Computed focus distance from LVDT	Link
FOCUS_LVDT_POS_NM	0x34	LVDT sensor position in nanometers	Link
TIMESTAMP_UNIX_EPOCH_SEC	0x38	Unix epoch timestamp in seconds	Link
TIMESTAMP_SUBSEC_NSEC	0x3C	Sub-second timestamp in nanoseconds	Link
VERSION	0x40	Subsystem version info	Link
RESERVED_PADDING0	0x44	Reserved zero padding	–
RESERVED_PADDING1	0x48	Reserved zero padding	–
METADATA_CRC32	0x4C	CRC32 of Metadata calculated by FPGA during capture	–

Table 47: Image Metadata Format

4 Farsight Camera Sensor Payload

FRASIGHT uses Sony IMX531 camera sensor for capturing non-earth images. The specification of the camera sensor is provided in this section.

4.1 IMX531 Camera Sensor Features

The IMX531-AAMJ-C is a diagonal 17.5 mm (Type 1.1) CMOS active pixel type solid-state image sensor with a square pixel array and 20.35 M effective pixels. This chip features a global shutter with variable charge-integration time [6].

- CMOS active pixel type dots
- Built-in timing adjustment circuit
- Horizontal and Vertical driver
- Serial communication circuit
- Global shutter function
- Embedded Clock frequency: 54 MHz
- Number of recommended recording pixels: 4504(H) × 4504(V) , approx. 20.28 M pixels
- Features supported in Farsight Avionics Firmware:
 - All-pixel scan mode
 - 12-bit pixel width
 - Max frame rate 64 fps for 8GB DDR4 and 70 fps for 16 GB DDR4 memory buffer
 - Variable-exposure duration via controled trigger pin
 - Monitor output for Exposure period via dedicated output pin
 - I/O interface SLVSEC 8 lane output
 - Data transfer rate at 4.752 Gbps
 - Programmable Gain Amplifier function
 - * 0 dB to 24 dB: Analog Gain (0.1 dB step)
 - * 24.1 dB to 48 dB: Analog Gain: 24 dB + Digital Gain: 0.1 dB to 24 dB (0.1 dB step)

4.1.1 IMX531 Camera Sensor Pixel Arrangement

The recommended recording pixels of the camera sensor is $4504 \times 4504 = 20.28$ Mega Pixel. The details of the pixel array is shown in Figure 3.

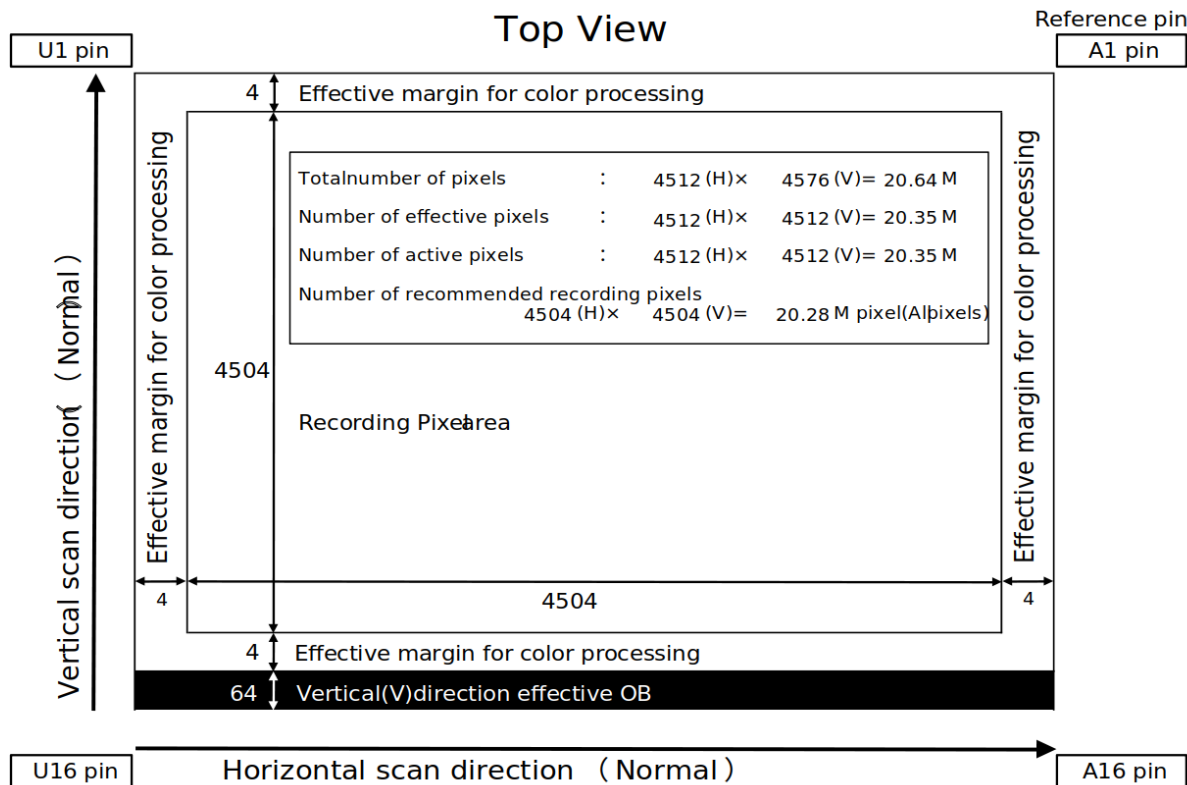


Figure 3: IMX531 SLVS-EC Pixel Array Image Drawing in All-pixel scan Mode [6]

4.1.2 IMX531 Camera Sensor Readout Modes

The IMX531 image sensor supports four different Readout modes. The readout modes are:

1. All Pixel scan
2. Vertical/Horizontal 1/2 subsampling
3. 2x2 FD binning
4. Region of interest (ROI) and Overlapped ROI

Readout modes are only valid during Operation modes. Meaning there won't be any reading during Standby mode. The pixel array structure of the above modes are described in this section.

In the current version of Farsight only the All Pixel scan mode is supported.

4.1.3 Farsight Packed Pixel Storage Format

Figure 4 illustrates how pixels are stored in the DDR4 memory space. Pixels are stored in ascending order, starting at the lowest memory address of the DDR4. In the current design, pixels are always 12 bits wide and stored in a packed format; therefore, unpacking to 16 bits is required before further processing of the image.

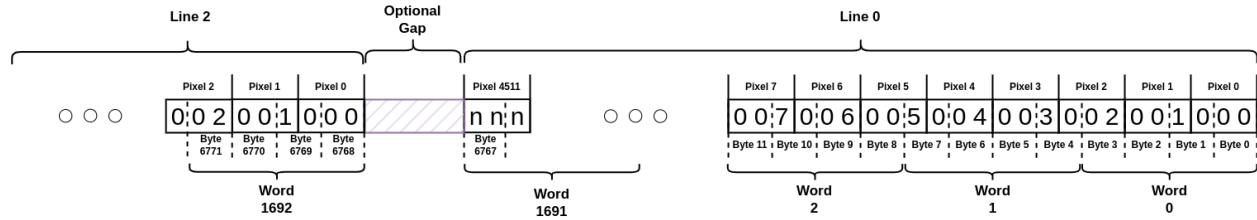


Figure 4: Farsight 12 bit Packed Pixel Frame Storage Format

The first line of the frame always contains the metadata generated by the FPGA system at the moment of frame capture. Details of the metadata format are discussed in the subsequent sections. The next 64 lines of the frame constitute the Vertical Effective Optical Black (OB) region, which is used for calibration and dark current compensation. The actual image data begins at line 65 and spans a total of 4512 lines. After the 64 lines of VOB there is a 4 line of effective margin for color processing at the beginning of the image and there is another 4 lines of effective margin for color processing at the end of the image. Each line of the frame consists of 4512 pixels, and since each pixel is 12 bits wide, the total size per line is $\frac{4512 \times 12}{8} = 6768$ bytes.

4.1.4 Farsight Packed Pixel UDP Output Format

The FPGA transmits image frames via UDP packets to a host computer. Each frame consists of 4512×4576 pixels, where the image data begins at line 65 (after the 64-line Optical Black region and 1 metadata line).

Farsight UDP Packet Structure Each UDP packet contains a 6-byte header followed by the pixel payload. Table 48 describes the packet header format.

Table 48: UDP Packet Header Structure

Field	Size (bytes)	Description
Start Marker	2	Packet delimiter
Line Sequence	2	Line number within the frame (big-endian)
Packet Sequence	2	Packet number within the line (big-endian)
Payload	Variable	Packed 12-bit pixel data

Each line is transmitted across 5 UDP packets (packets 0–4), resulting in a total of $4576 \times 5 = 22,880$ packets per frame.

Camera Sensor Row Format Each row contains 4512 pixels in packed 12-bit format, yielding $\frac{4512 \times 12}{8} = 6768$ bytes of pixel data. Additionally, each row includes 16 bytes of padding for memory alignment purposes, resulting in a total row size of $6768 + 16 = 6784$ bytes.

The expected total frame size can be calculated as:

$$\text{Frame Size} = \text{Height} \times (\text{Packed Row Bytes} + \text{Padding}) = 4576 \times (6768 + 16) = 31,039,484 \text{ bytes} \quad (1)$$

Camera Sensor 12-bit Packed Pixel Unpacking Instruction Pixels are stored in a packed format where every 12 bytes (96 bits) encode 8 pixels. The unpacking process interprets the data as three 32-bit big-endian words (W_0 , W_1 , W_2) and extracts the 8 pixels using bitwise operations as shown in Table 49.

To convert 12-bit pixel values to 16-bit format for standard image processing applications, the pixel values are shifted left by 4 bits:

$$P_{16} = P_{12} \ll 4 \quad (2)$$

Image Data	Checksum placeholder	Start Marker	Line Sequence	UDP Packet Sequence
00 24 9b 8b ee fb 00 04	a3 12	34 56	08 00	45 00
05 ce 00 01 00 00 40 11	40 d6	0a 65	0f c0	0a 65
0f bf 04 d2 89 31 05 ba	00 00	50 4b	00 10	00 02
30 00 04 50 00 06 70 00	08 90	00 0a	b0 00	0c d0
00 0e f0 00 10 10 01 12	30 01	14 50	01 16	70 01
18 90 01 1a b0 01 1c d0	01 1e	f0 01	20 10	02 22
30 02 24 50 02 26 70 02	28 90	02 2a	b0 02	2c d0
02 2e f0 02 30 10 03 32	30 03	34 50	03 36	70 03
38 90 03 3a b0 03 3c d0	03 3e	f0 03	40 10	04 42
30 04 44 50 04 46 70 04	48 90	04 4a	b0 04	4c d0
04 4e f0 04 50 10 05 52	30 05	54 50	05 56	70 05
58 90 05 5a b0 05 5c d0	05 5e	f0 05	60 10	06 62

Figure 5: Farsight 12 bit Packed UDP Output Format

Table 49: 12-bit Packed Pixel Unpacking Formulas

Pixel	Extraction Formula
P_0	$W_0 \wedge 0xFFF$
P_1	$(W_0 \gg 12) \wedge 0xFFF$
P_2	$((W_0 \gg 24) \vee (W_1 \ll 8)) \wedge 0xFFF$
P_3	$(W_1 \gg 4) \wedge 0xFFF$
P_4	$(W_1 \gg 16) \wedge 0xFFF$
P_5	$((W_1 \gg 28) \vee (W_2 \ll 4)) \wedge 0xFFF$
P_6	$(W_2 \gg 8) \wedge 0xFFF$
P_7	$(W_2 \gg 20) \wedge 0xFFF$

Payload Computer Receiver Configuration Requirements To reliably receive high-bandwidth UDP image data without packet loss, the host system must be configured with adequate UDP receive buffer sizes. The recommended minimum buffer size is 256 MB. On Linux systems, this can be configured using:

```
sudo sysctl -w net.core.rmem_max=268435456
sudo sysctl -w net.core.rmem_default=268435456
```

To make these changes persistent across reboots, add the following lines to `/etc/sysctl.conf`:

```
net.core.rmem_max = 268435456
net.core.rmem_default = 268435456
```

Farsight UDP Packet Loss Handling Due to the nature of UDP transmission, packets may occasionally be lost or arrive out of order. The receiver software implements the following strategies to handle these scenarios:

- **Packet Reordering:** Packets are buffered and written in the correct sequence based on line and packet sequence numbers.

- **Timeout Mechanism:** If a packet does not arrive within a configurable wait time (default: 10 ms), the missing data is filled with zeros.
- **Frame Timeout:** If a complete frame is not received within the frame timeout period (default: 2 seconds), the frame is finalized with any missing packets filled with zeros.

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5 Farsight Hardware Architecture

5.1 Farsight Physical Interfaces

Figure 6 shows the physical interfaces of the FARSIGHT Payload system.

The FARSIGHT system connects to external components through the following interfaces:

- **External Interfaces:** The system interfaces with the Xiphos Q8JS Payload Computer via multiple communication channels:
 - **RS422:** Primary TMTC (Telemetry and Telecommand) serial interface.
 - **1GigE (x2):** Dual Gigabit Ethernet connections for backup and primary data transfer.
 - **PCIe x2 Gen2:** High-speed PCIe interface for primary data transfer.
 - **SPI:** Used for reflash operations.
 - **PPS:** Pulse Per Second timing signal source.
 - **GPIOs (4x):** External trigger input (EXT.TRIG), interrupt output (INTR), and general-purpose I/O signals.
- **ProASIC3 FPGA:** Handles power sequencing and system monitoring.
- **PolarFire FPGA (MPF500TS-FC1152M):** The main processing unit that interfaces with:
 - **64 Mb Microchip RT NVM:** Non-volatile memory connected via QSPI for configuration storage.
 - **Micron DDR4 (24 GB):** Frame buffer memory (18x MT40A2G7AG-062E AUT:F) for image storage.
- **Sony IMX531-AAMJ-C Sensor:** Connected to the PolarFire FPGA via:
 - **8-Lane SLVS-EC:** High-speed data interface running at 4.752 Gbps/lane.
 - **SPI:** Sensor control interface.
 - **XTRG1:** Exposure trigger GPIO.
 - **XCLR:** Settings clear GPIO.
 - **TOUT0:** Exposure timing knowledge GPIO.
- **FARSIGHT Optical Assembly:** Includes 13x thermistors for temperature monitoring and 7x heaters for thermal control.
- **Focus Mechanism:** Controlled by the PolarFire FPGA with LVDT feedback for position sensing.
- **Heater and Thermistor:** Additional thermal management components with direct connection to the electrical interface.

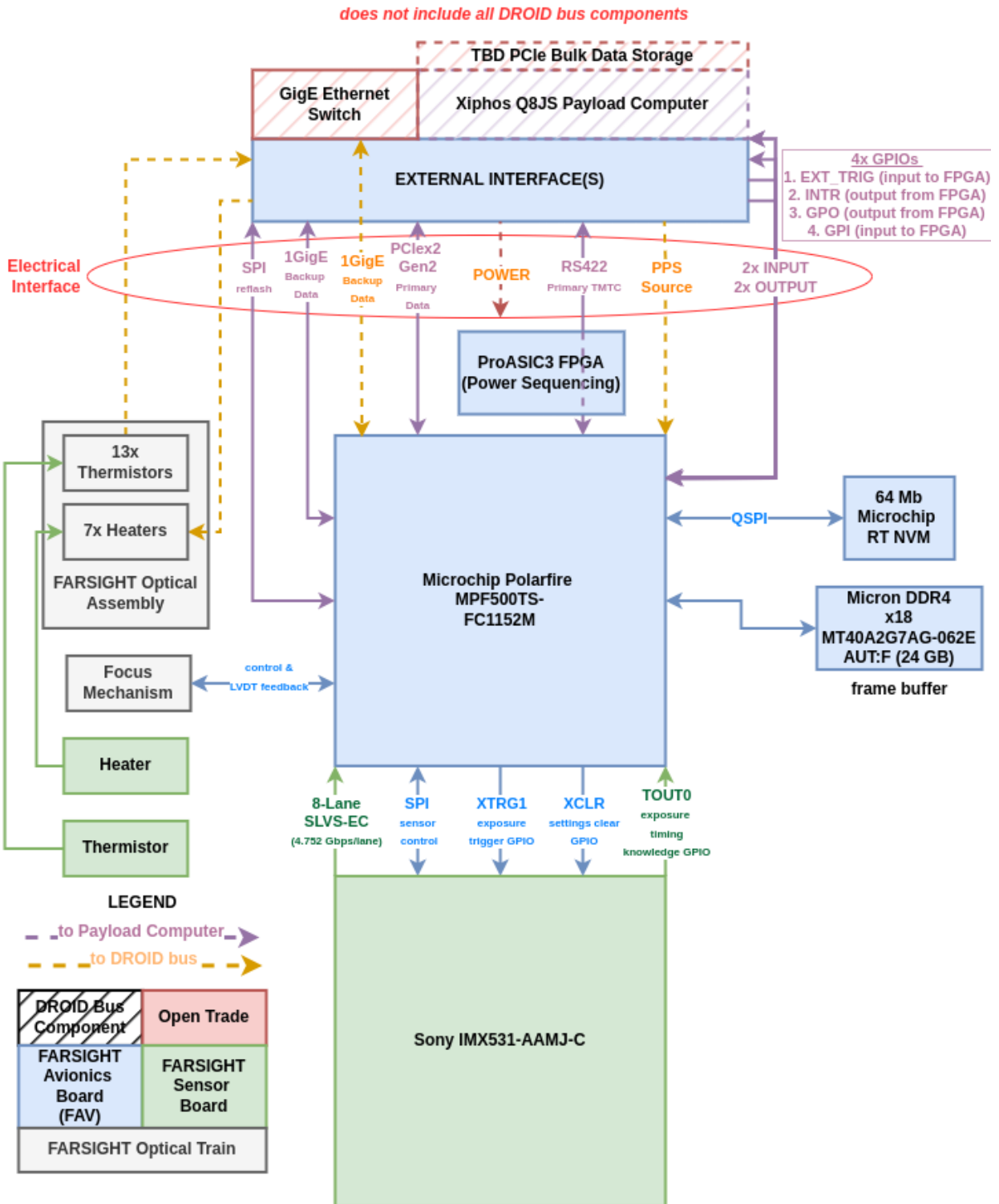


Figure 6: Farsight Physical Interfaces

5.2 Farsight Highlevel Hardware Design

Figure 7 shows the high level architecture of FARSIGHT Payload system.

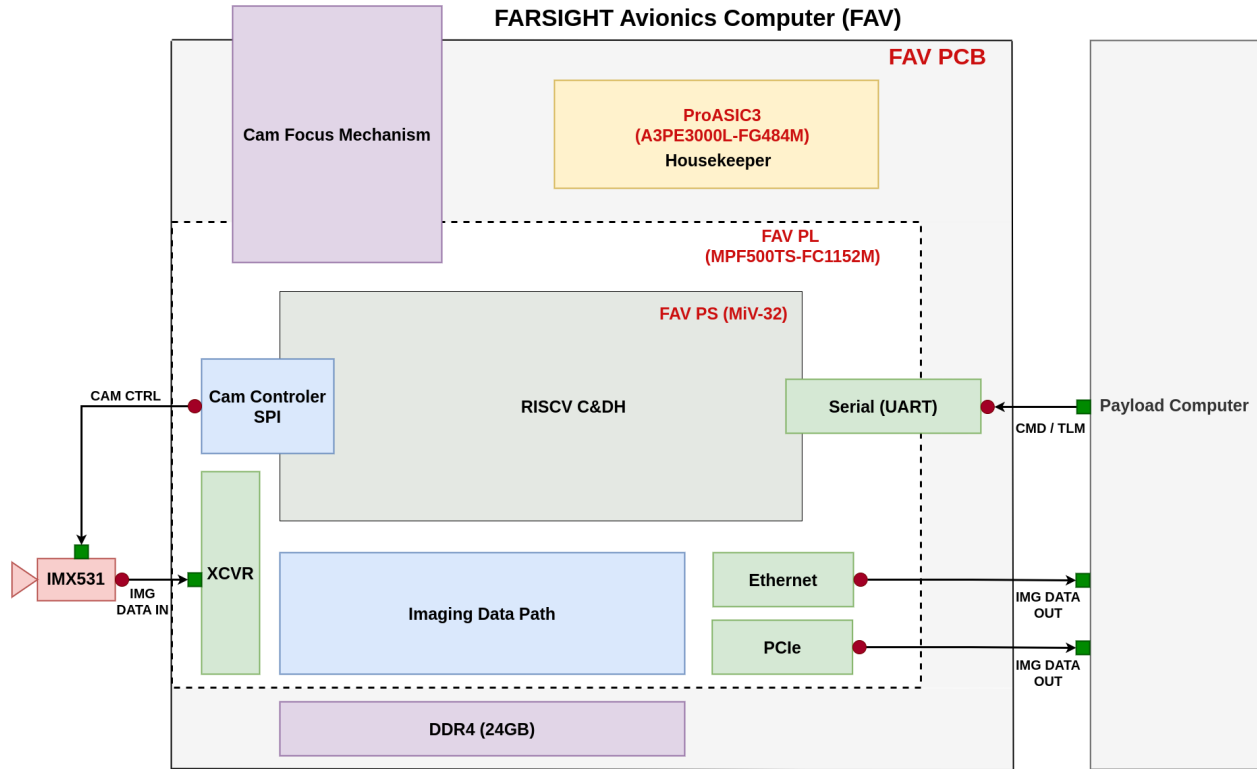


Figure 7: Farsight Avionics Architecture High Level Blocks

The Farsight Avionics Computer (FAV) is the interface between the Payload Computer and the Camera sensor (IMX531). The FARSIGHT electronics reside on the FAV PCB which consists of two main FPGA chips and supporting components:

- **ProASIC3 (A3PE3000L-FG484M) - Housekeeper FPGA:** The Housekeeper FPGA monitors power fluctuations and performs latchup protection in case of single-event latchup (SEL) issues caused by radiation.
- **PolarFire (MPF500TS-FC1152M) - FAV PL:** The PolarFire FPGA serves as the main control interface and contains two primary subsystems:
 - **FAV PS (MiV-32):** The Processing System hosts the RISC-V MiV-32 softcore processor which acts as the main Command and Data Handling (C&DH) unit. It is responsible for receiving and processing telecommands from the Payload Computer and responding with telemetry information.
 - **Imaging Data Path:** The programmable logic section handles high-speed image data processing and frame buffer management.

The main Telemetry and Telecommand (TMTC) interface between Farsight Avionics and the Payload Computer is a Serial UART interface backed by RS422 differential signaling for reliable CMD/TLM communication.

The Camera Sensor interface to the system consists of two main paths: the control path and the data path.

- **Control Path (CAM CTRL):** The Cam Controller uses a SPI interface that allows configuration of the IMX531 camera sensor via SPI protocol.

- **Data Path (IMG DATA IN):** The data path interface consists of 8-lane high-speed SERDES lanes running at 4752 Mbps/lane connected to the transceivers (XCVR) of the PolarFire FPGA. The image data passes through the Imaging Data Path pipeline and is stored in frame buffers allocated in the external 24GB DDR4 memory.

Additionally, the FAV includes a Camera Focus Mechanism for controlling the camera lens focus.

For image data output (IMG DATA OUT), there are two main options to transfer data to the Payload Computer:

- **Ethernet:** The 1 Gigabit Ethernet interface supports up to 3 fps using User Datagram Protocol (UDP).
- **PCIe:** The PCIe endpoint interface supports 5 Gbps data rate using PCIe Gen2.

5.2.1 Imaging Control Path

Figure 8 shows the high-level view of the Farsight Avionics imaging control path.

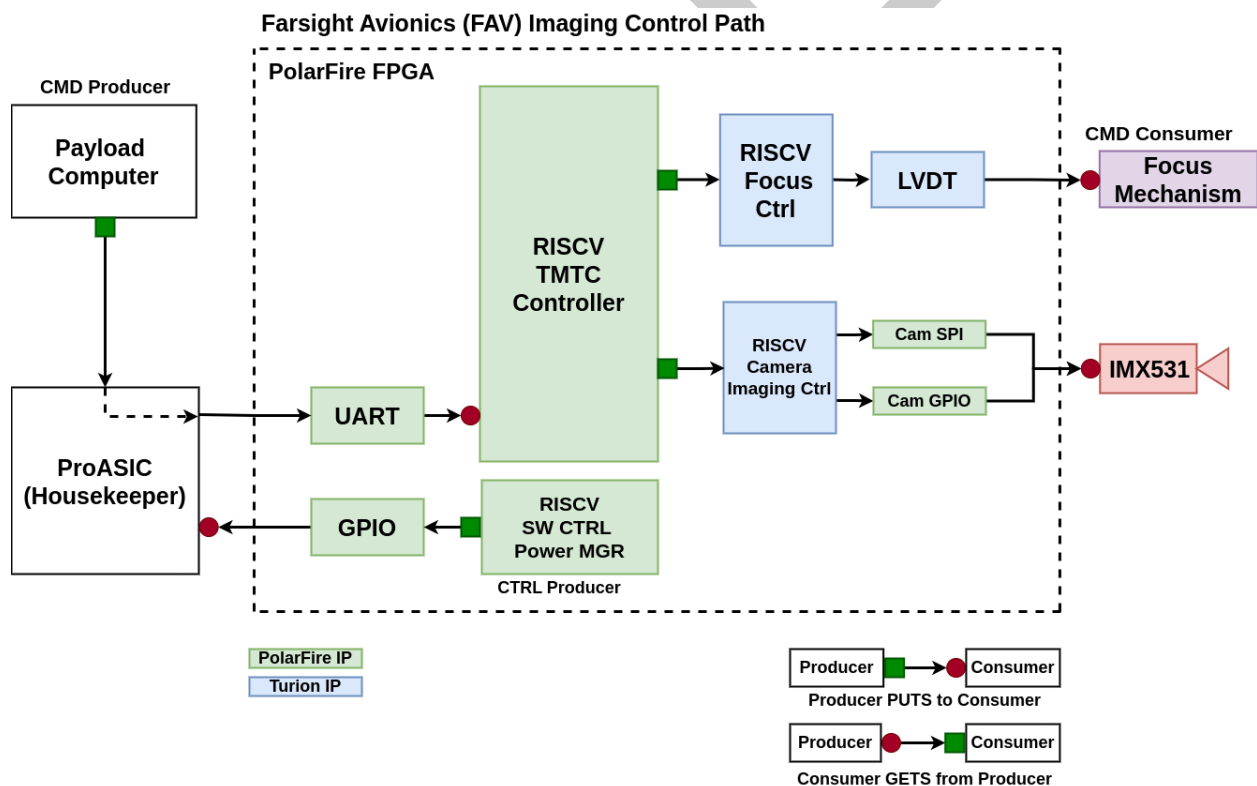


Figure 8: Farsight Avionics Imaging Control Path

The Payload Computer sends telecommands to the FAV through a communication path that passes through the Housekeeper FPGA. Under nominal conditions, the Housekeeper FPGA operates as a pass-through for this communication link as long as the PolarFire FPGA is powered on. In the event that the PolarFire FPGA fails to power on, the Housekeeper FPGA assumes control of this path and transmits a beacon signal back to the Payload Computer indicating which power region has failed.

During nominal operation, telecommands from the Payload Computer are received by the RISC-V processor in the PolarFire FPGA via the serial port interface. The RISC-V firmware parses these commands and dispatches them to the appropriate target subsystem. The imaging control path consists of the following controllable subsystems:

- **Camera Control Module:** The camera control module consists of two main components:

- **Camera SPI Interface:** Provides configuration access to the IMX531 camera sensor registers via SPI protocol.
- **Trigger Module:** Generates trigger pulses to initiate image capture based on user-specified exposure settings and frame rate (frames per second) parameters.
- **Focus Mechanism Controller:** Enables precise positioning of the camera focus mechanism through telecommands to achieve optimal image focus.
- **Housekeeper Interface Module:** Provides the RISC-V firmware with the ability to monitor system status and issue commands to the Housekeeper FPGA for controlling various power regions within the system.

5.2.2 Imaging Data Path

Figure 9 shows the high level view of Farsight Avionics Imaging Data Path.

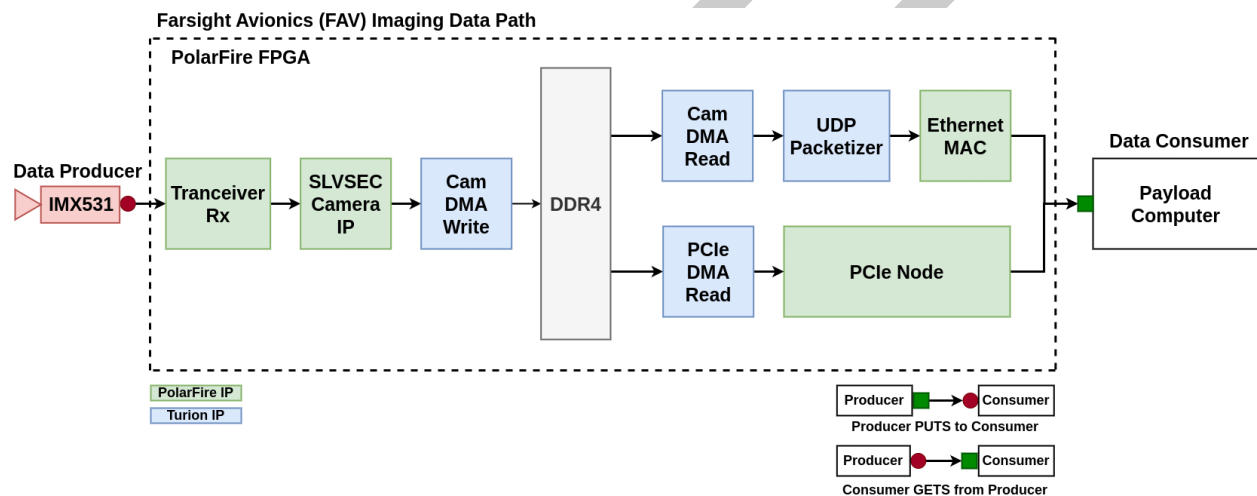


Figure 9: Farsight Avionics Imaging Data Path

The imaging data path illustrates how image data flows from the IMX531 sensor to the Payload Computer. The IMX531 sensor acts as the data producer, transmitting image data through the Transceiver Rx block into the SLVSEC Camera IP within the PolarFire FPGA. The Camera DMA Write module then stores the incoming image frames into DDR4 memory.

From DDR4 memory, the system supports two parallel data paths to the Payload Computer (data consumer):

- **Ethernet Path:** The Cam DMA Read module retrieves image data from DDR4, which is then packetized by the UDP Packetizer and transmitted through the Ethernet MAC to the Payload Computer.
- **PCIe Path:** The PCIe DMA Read module retrieves image data from DDR4 and transfers it through the PCIe Node directly to the Payload Computer.

5.2.3 Focus Mechanism

Figure 10 shows the focus mechanism control system architecture.

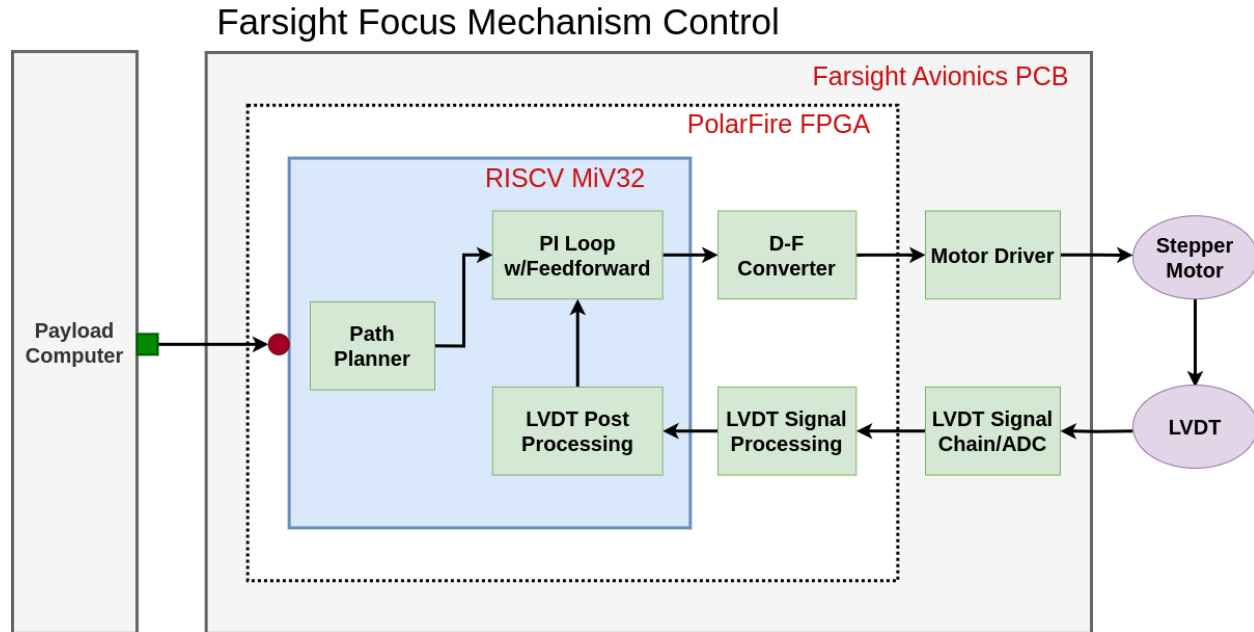


Figure 10: Farsight Focus Mechanism Control

The focus mechanism control loop operates as follows: Upon receiving a focus command from the Payload Computer, the RISC-V MiV32 processor initiates path planning. If the commanded setpoint is too close to the current position, the Path Planner will first move the sensor away from the target before approaching the setpoint to avoid stiction issues.

The Path Planner generates updated velocity and position commands which are fed to the PI Loop with Feedforward. The PI loop employs velocity feedforward to improve motion profile tracking accuracy. Position feedback is provided by a Linear Variable Differential Transformer (LVDT) sensor, whose signal passes through the LVDT Signal Chain/ADC for analog-to-digital conversion, followed by LVDT Signal Processing and LVDT Post Processing blocks before being fed back to the PI loop.

The PI loop output is converted by the D-F Converter and sent to the Motor Driver, which commands the Stepper Motor to adjust the focus mechanism position. The motor is stopped once the desired position is reached, or if the position error exceeds a specified threshold in microns.

5.2.4 Hardware Power Protection

Farsight Avionics Housekeeper consists of two FPGA chips for managing and monitoring power status of the system. The ProASIC3 FPGA and PolarFire FPGA. The ProASIC3 FPGA Housekeeper is responsible for controlling and monitoring power to various subsystems on the Farsight Avionics board. It manages 33 power sources organized into 11 power regions, each with dedicated state machines for sequencing and fault detection. During normal operation, the ProASIC3 operates in pass-through mode, routing UART traffic directly between the PolarFire and the payload bus. If a critical failure occurs and the PolarFire is powered down, the ProASIC3 takes control of the UART and broadcasts a failure status message approximately once per second. The message format is shown in Table 66.

6 Concept of Operations (ConOps)

This section describes the operational procedures for the Farsight Avionics system. Each ConOps provides step-by-step instructions for specific operational scenarios.

6.1 Nominal Image Capture (Without Focus Mechanism)

This procedure describes the nominal image capture sequence for Farsight Avionics without the focus mechanism engaged. The procedure is divided into phases, each containing sequential steps that must be completed before proceeding to the next phase.

6.1.1 Phase 1: Power On

1. Power ON the Farsight Avionics system.
2. Confirm there is no power failure message from the ProASIC3 Housekeeper within 0–500 ms after power on.

6.1.2 Phase 2: Preliminary TMTC Functionality Check

3. Read `SENSOR_STATE_SYS_REG` via `REG_READ_SYS_CMD` and verify the return value is `LP_MODE`.
4. Read `VERSION_SYS_REG` via `REG_READ_SYS_CMD` and verify the expected firmware version values are returned.
5. Send `ECHO_SYS_CMD` and verify the payload is correctly echoed back.

6.1.3 Phase 3: Power Region Status Check

6. Read `HOUSEKEEPER_STATUS_SYS_REG` via `REG_READ_SYS_CMD` and verify the following power region status bits:
 - Bit 3: 16GB DDR4 power status
 - Bit 2: 8GB DDR4 power status
 - Bit 6: ETH1 PHY power status

6.1.4 Phase 4: Register Write/Read Test via TMTC

7. Perform `REG_WRITE_SYS_CMD` to `NOOP_SYS_REG` and read back the register via `REG_READ_SYS_CMD` to verify read/write functionality.

6.1.5 Phase 5: System Parameter Configuration

8. Set real time clock based on the next PPS signal using `SET_TIME_SYS_CMD`.
9. Set the network parameters via `REG_WRITE_SYS_CMD`:
 - Source MAC address
 - Destination MAC address
 - Source/Destination IP addresses
 - UDP port number

6.1.6 Phase 6: Ethernet Functionality Check

10. Ping ETH1 interface to verify network connectivity.

6.1.7 Phase 7: IMX Sensor Power On and Initialization

11. Write `IDLE_MODE` to `SENSOR_STATE_SYS_REG` via `REG_WRITE_SYS_CMD` to change mode to `IDLE_MODE`. This powers on the IMX531 sensor and configures all required initial registers to place the sensor into a functional standby state.
12. Read the sensor temperature via `REG_READ_SYS_CMD` on register `SENSOR_TEMP_RAW_SYS_REG` to verify sensor communication.

6.1.8 Phase 8: Image Capture Configuration

13. Write 0 to `BUFF_WRITE_INDEX_SYS_REG` via `REG_WRITE_SYS_CMD` to reset the frame buffer write pointer.
14. Set the exposure time via `REG_WRITE_SYS_CMD` on register `SENSOR_EXPO_USEC_SYS_REG`.
15. Set the black level offset via `REG_WRITE_SYS_CMD` on register `SENSOR_BLO_SYS_REG`.
16. Set the sensor gain via `REG_WRITE_SYS_CMD` on register `SENSOR_GAIN_SYS_REG`.
17. Write the number of frames to capture (e.g., 1) to register `SENSOR_NUM_FRM_TO_CAPTURE_SYS_REG` via `REG_WRITE_SYS_CMD`.
18. Write `TRIG_ON_CMD` to register `SENSOR_TRIG_MODE_SYS_REG` via `REG_WRITE_SYS_CMD` to set the trigger mode to command trigger.
19. Arm the sensor by writing `ARMED_MODE` to `SENSOR_STATE_SYS_REG` via `REG_WRITE_SYS_CMD`.

6.1.9 Phase 9: Image Capture Trigger

20. Start image capture via `START_ACQ_SYS_CMD`.
21. Poll `BUFF_WRITE_INDEX_SYS_REG` via `REG_READ_SYS_CMD`. When the value increments, the image capture is complete.

6.1.10 Phase 10: Image Transfer via UDP

22. Initiate the UDP receiver script on the host computer.
23. Transfer the image frame via `XFER_BUFF_SYS_CMD` with parameters:
 - `image_index`: 0
 - `num_images`: 1
24. Extract metadata from the received frame and confirm the recorded values (timestamp, frame number, exposure settings).
25. Convert the received raw image to the desired format (PNG, TIFF).
26. (Optional) Compare the generated image against a known golden reference image to verify transfer integrity.

6.1.11 Phase 11: Power Down

27. Power down the system via the Housekeeper power control interface.

6.2 Nominal Focus Mechanism Operation

TO BE COMPLETED

6.3 Fault Handling Operation

TO BE COMPLETED

DRAFT

7 FPGA Design Document

This document describes the FARSIGHT FPGA architecture in details.

7.1 Farsight PolarFire FPGA Chip

Category	Specification	Value
FPGA Fabric	Logic Elements (4LUT + DFF)	481K
	Math Blocks (18 × 18 MACC)	1480
	LSRAM Blocks (20 Kb)	1520
	uSRAM Blocks (64 × 12)	4440
	Total RAM (Mb)	33
	uPROM (Kb)	513
	User DLLs/PLLs	8 each
High-Speed I/O	250 Mbps – 12.7 Gbps Transceiver Lanes	24
	PCIe Gen 2 Endpoints/Root Ports	2
Total I/O	Total User I/O	584

Table 50: MPF500T FPGA Specifications [1]

In this section, we will review the main components of the FAV Hardware (FAV-HW). The HW will consist of 32GB of DDR4 memory as volatile memory storage. The DDR4 space will be solely used for buffering captured images from the camera. The HW also consists of a radiation tolerant Non-Volatile memory of size 64 MB with SPI interface. The external NVM will be used for storing FAV parameters and critical TLM logs. There will be also two radiation tolerant Ethernet PHY VSC8541RT used as redundancy backup. The HW also will provide 148.5 MHz Ref clock for the XCVR usage and 50 MHz fabric clock for the Programmable Logic clocking. The FARSIGHT target FPGA chip is Microchip PolarFire radiation tolerant MPF500T. Table 50 shows the MPF500T specification.

7.2 FPGA Architecture

The FARSIGHT FPGA consists of the following main groups of design architecture:

1. **Camera control and data receiving**
2. **Data write and read to the DDR4 memory space pipeline**
3. **UDP Data export components**
4. **Ethernet MAC**
5. **Housekeeper**
6. **Focus Mechanism**
7. **Softcore RISC-V MiV32 processing system**

Each of these components will be discussed in detail in the following sections.

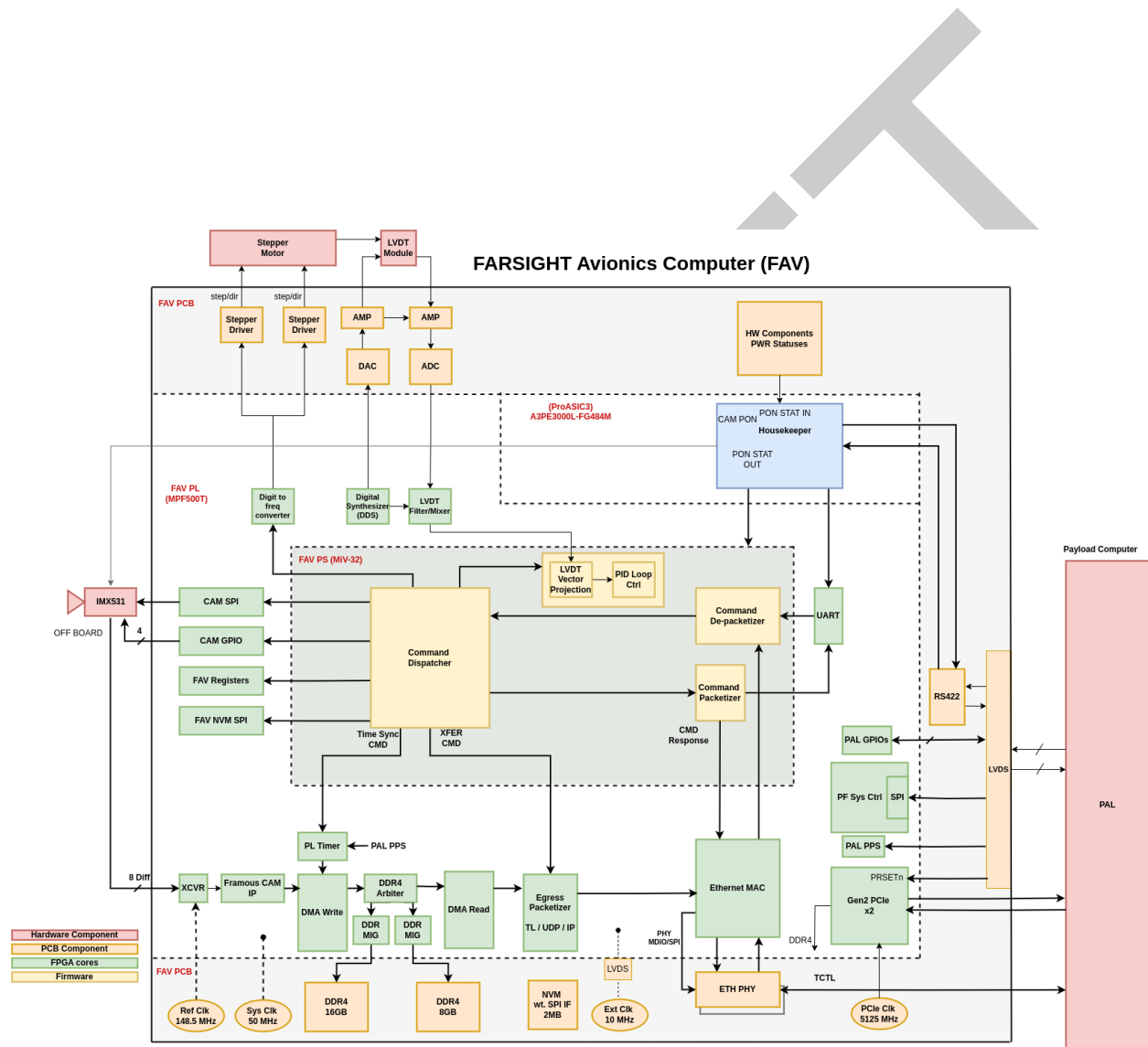


Figure 11: Farsight Avionics Computer FPGA Architecture

7.3 Camera Control and Data Receiving Architecture

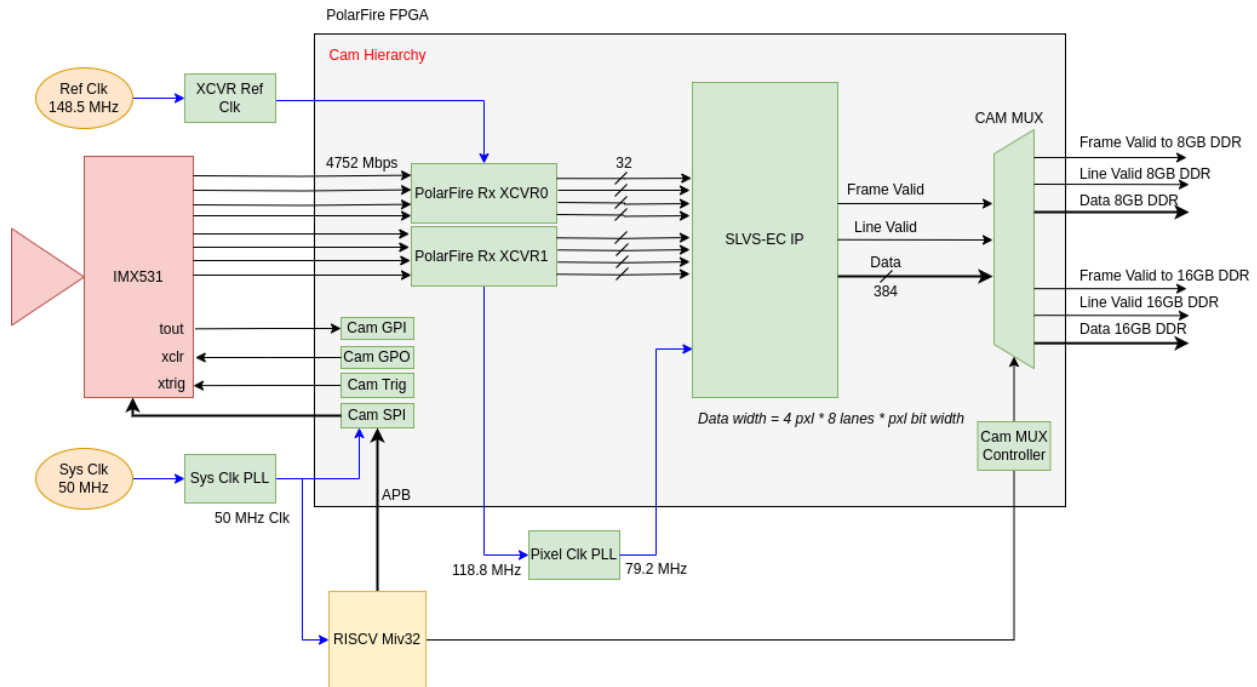


Figure 12: Farsight Avionics Camera Control Architecture

The camera control architecture is depicted in Figure 12. This subsystem handles high-speed image data reception from the camera sensor using the SLVS-EC (Scalable Low Voltage Signaling with Embedded Clock) interface.

7.3.1 High-Speed Transceiver Interface

The design utilizes two PolarFire XCVR (transceiver) instances, each supporting 4 lanes, for a total of 8 differential SLVS-EC lanes. The transceivers are configured with the following parameters:

- CDR reference clock: 148.5 MHz
- RX data rate: 4.752 Gbps per lane
- PCS fabric interface width: 40 bits
- Rx Clock Out Freq: 118.8 MHz

The transceivers operate in PMA (Physical Medium Attachment) mode, providing raw serialization deserialization. Each transceiver lane outputs 40-bit encoded data to a CorePCS instance that performs 8B/10B decoding and word alignment using K28.5 comma detection (single comma required for alignment). The CorePCS decodes the 40-bit input (4×10 -bit symbols) into 32-bit data (4×8 -bit bytes). The decoded data then passes through a `word_alignment` module that performs byte-order swapping to correct endianness. The SLVS-EC RX IP aggregates data from all 8 lanes into a 384-bit output bus (8 lanes $\times$ 12-bit pixel depth $\times$ 4 pixels per lane) along with frame and line valid signals.

7.3.2 Clock Domains

The camera control subsystem operates across multiple clock domains:

- **APB Clock (50 MHz):** Used for register configuration and control interfaces.
- **CDR Reference Clock (148.5 MHz):** Provides the reference for the transceiver CDR (Clock and Data Recovery) circuits.
- **Pixel Reference Clock (118.8 MHz):** Recovered clock from Lane 0, used as the reference for downstream pixel processing.
- **Pixel Fabric Clock (79.2 Mhz):** Clock domain for the SLVS-EC RX IP and the camera MUX logic. See below for the clock calculation formula.

PLL for Pixel Clock Generation The pixel fabric clock is derived from the recovered lane clock using the following formula [5]:

$$f_{\text{pixel}} = \frac{f_{\text{LANE0_RX_CLK}} \times \text{NUM_LANES}}{\text{PIXEL_WIDTH}} \quad (3)$$

For this design:

$$f_{\text{pixel}} = \frac{118.8 \text{ MHz} \times 8}{12} = 79.2 \text{ MHz} \quad (4)$$

7.3.3 Camera MUX

The `cam_mux` module routes the received camera data to one of two DDR4 memory spaces (8 GB or 16 GB). It operates in the pixel clock domain and uses a state machine to ensure frame-aligned switching between destinations. This means if the IMX is already streaming data and the MUX is enabled in the middle of a frame transfer the MUX won't allow the data to move out until the start of the next valid camera frame. The MUX is controlled via APB-accessible GPIO registers that set the enable and select signals.

7.3.4 Camera MUX APB Register

The camera MUX is controlled through a CoreGPIO instance accessible via the APB bus. Table 51 describes the GPIO output bits used for MUX control.

Table 51: Camera MUX GPIO Control Bits

Bit	Name	Default	Description
0	MUX_ENABLE	0	MUX enable control (0 = disabled, 1 = enabled)
1	MUX_OUTPUT_SELECT	0	Memory destination select (0 = DDR4 8GB, 1 = DDR4 16GB)

7.3.5 Camera GPIO Interface

Camera GPIO interface consists of 3 main signals:

- **XCLRn:** IMX sensor active low reset.
- **XTRIG:** Signal to the camera sensor for frame capture control and exposure time.
- **TOUT:** Signal from IMX sensor indicating the on going exposure duration signal.

Each of these signals are specified in the next sections.

7.3.6 Camera Reset (XCLRn)

This signal resets the IMX register to the default values. Hold XCLR at Low level for 500 ns or more after all the power supplies have finished rising [6].

7.3.7 Camera Trigger (XTRIG)

The `cam_trig` module generates the XTRIG signal to the camera sensor for frame capture control signaling and exposure time. It is software-configurable via an APB register interface and allows control over trigger timing and frame capture count.

Camera Trigger Functional Overview The camera trigger IP operates as a finite state machine with the following states:

- **OFF**: Initial state, waiting for enable signal
- **IDLE**: Enabled and ready to accept configuration, waiting for start command
- **XTRIG_LOW_PERIOD**: Driving XTRIG low for the configured duration
- **XTRIG_HIGH_PERIOD**: Driving XTRIG high for the remainder of the frame period
- **DONE**: Capture sequence complete, returns to IDLE

The XTRIG signal timing is controlled by three parameters: the low pulse duration, the total frame capture period, and the number of frames to capture. Setting `frame_capture_amount` to 0 enables continuous (infinite) triggering until the module is disabled.

Camera Trigger Timing Configuration All timing parameters are specified in microseconds and internally converted to clock cycles based on the `CLOCK_FREQ_MHZ` parameter (default: 50 MHz). The frame period defines the time between consecutive trigger pulses:

$$T_{\text{high}} = T_{\text{frame}} - T_{\text{low}} \quad (5)$$

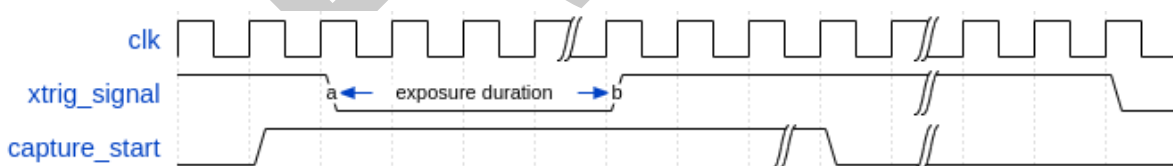


Figure 13: Farsight Avionics Camera Trigger Waveform

Camera Trigger Usage Example To capture 100 frames at 50 fps with a 10 μs trigger pulse:

1. Write 10 to XTRIG_LOW_TIME Reg (0x00)
2. Write 20000 to FRAME_CAPTURE_TIME Reg (0x04) for ~50 fps
3. Write 100 to FRAME_CAPTURE_AMOUNT Reg (0x08)
4. Write 1 to START (0x0C) Reg to begin capture

7.3.8 Camera Trigger APB Register Map

Table 52 shows the APB register interface for the camera trigger module.

Table 52: Camera Trigger APB Register Map

Offset	Name	Width	Access	Description
0x00	XTRIG_LOW_TIME	15 bits	R/W	Duration of XTRIG low pulse in microseconds (0–32767 μ s)
0x04	FRAME_CAPTURE_TIME	20 bits	R/W	Total frame capture period in microseconds (0–1048575 μ s)
0x08	FRAME_CAPTURE_AMOUNT	10 bits	R/W	Number of frames to capture (0 = infinite, 1–1023 = finite count)
0x0C	START	1 bit	R/W	Write 1 to start capture sequence; auto-clears on completion

7.3.9 Camera Exposure ACK Signal (TOUT)

The IMX TOUT signal is an acknowledgement signal for the period of the current running global shutter exposure.

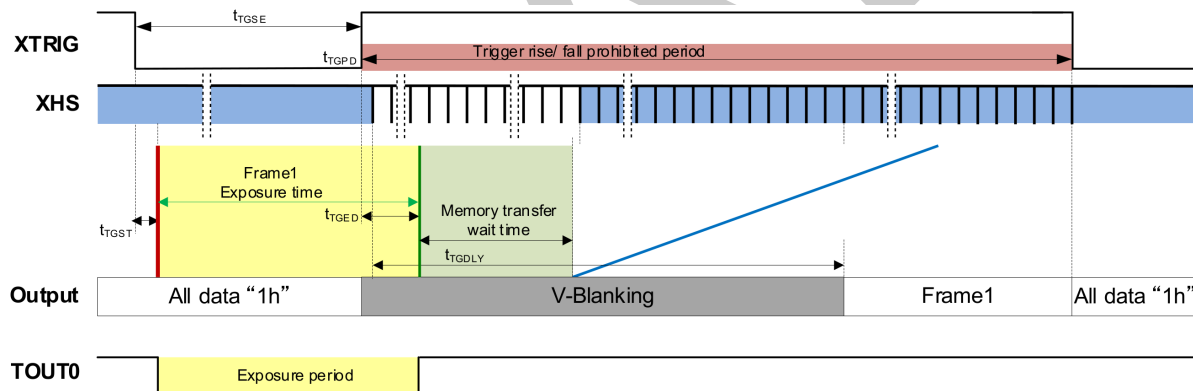


Figure 14: IMX TOUT Timing Diagram. See [6]

The parameters in the Figure 14 are described in the following:

1. TGST: Global Shutter Start Delay
2. TGED: Global Shutter End Delay
3. TGSE: Global Shutter Pulse width
4. TGPD: Global Shutter Next trigger rise prohibited period
5. TGDLY: Global Shutter Data output delay
6. TGES: Global Shutter Next trigger fall prohibited period

7.3.10 Camera SPI

The camera SPI interface provides register access to the IMX sensor for configuration and status readback. The design uses a Microchip CoreSPI instance configured as an SPI master with an APB slave interface.

SPI Configuration All the SPI output signals are kept in High-Z until the Housekeeper `cam_pwr_status` is high. Table 53 summarizes the CoreSPI configuration parameters.

Table 53: Camera SPI Configuration Parameters

Parameter	Value
APB Data Width	32 bits
SPI Mode	Motorola
Frame Size	8 bits
FIFO Depth	32 entries
Clock Divider	124
Motorola Mode	Mode 3 (CPOL=1, CPHA=0)
Bit Order	MSB first
SSEL Behavior	Keep active between frames
SSEL Polarity	Active low

Timing Characteristics In Motorola Mode 3, the SPI clock (SCK) idles high and data is sampled on the leading (falling) edge of the clock. The SPI clock frequency is derived from the APB clock divided by the clock divider value:

$$f_{\text{SPI}} = \frac{f_{\text{APB}}}{2 \times \text{CLK_DIV}} = \frac{50 \text{ MHz}}{2 \times 124} \approx 201.6 \text{ kHz} \quad (6)$$

7.4 Camera Data Pipeline Architecture

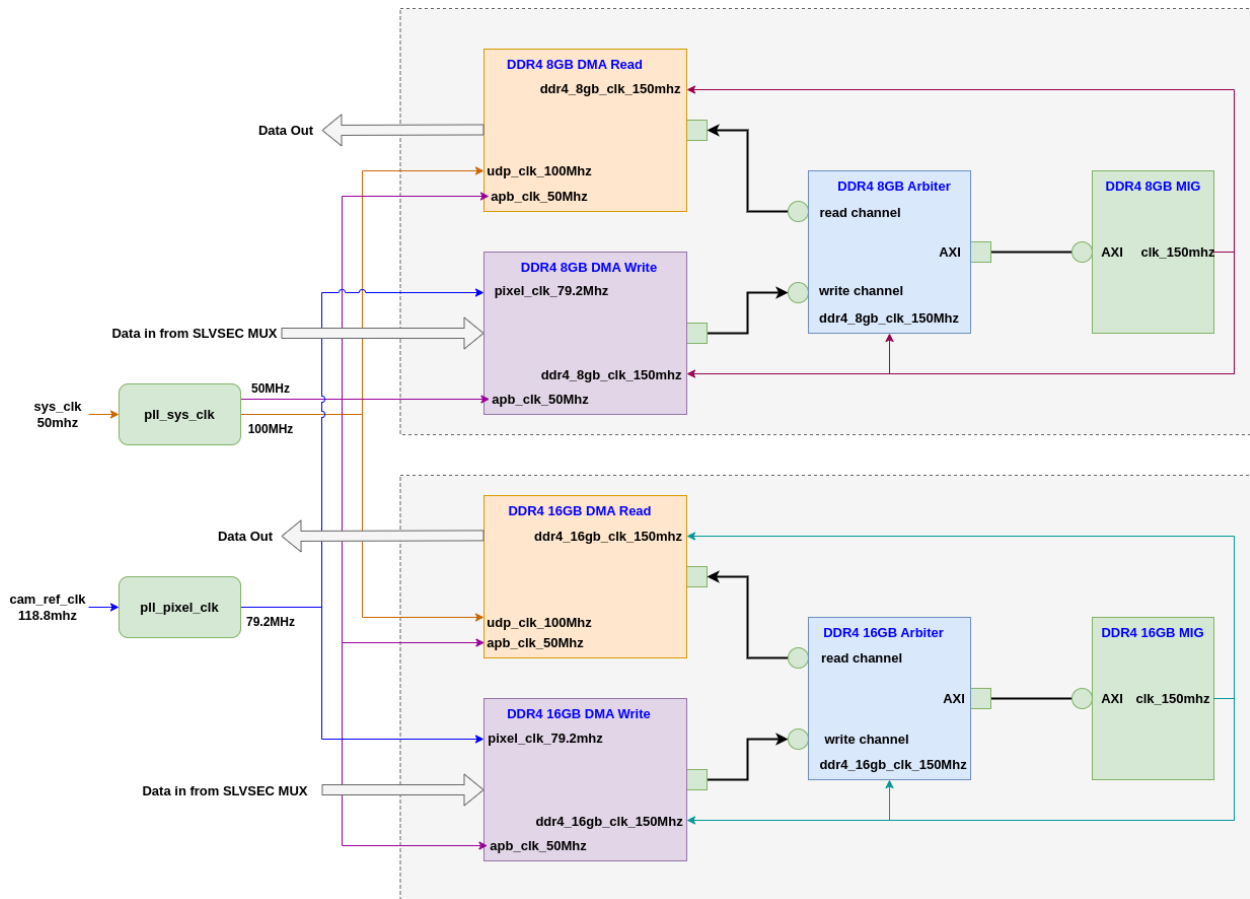


Figure 15: Farsight Avionics Data Pipeline Architecture

FARSIGHT Hardware consists of 2 DDR4 banks of 8 GB and 16 GB. To interface with DDR4 banks the following 4 main components depicted in Figure 15 are used:

1. DDR4 MIG (Memory Interface Generator) (8 GB and 16 GB)
2. Arbitrator
3. DMA Write
4. DMA Read

7.4.1 DDR4 MIG

The FARSIGHT design uses two PolarFire DDR4 Memory Interface Generator (MIG) instances to interface with the external DDR4 memory banks. Each MIG instance includes a DDR4 PHY, memory controller, and an AXI4 interface wrapper (CoreDDR_LiteAXI) for fabric connectivity.

DDR4 Memory Interface Configuration Table 54 summarizes the configuration parameters for both DDR4 memory interfaces. The 16 GB interface provides higher bandwidth with a wider data path, while the 8 GB interface uses a narrower configuration.

Table 54: DDR4 MIG Configuration Parameters

Parameter	16 GB	8 GB
DDR4 DQ Width	64 bits	32 bits
AXI Data Width	512 bits	256 bits
AXI Address Width	39 bits	38 bits
AXI ID Width	4 bits	4 bits
DDR Clock Frequency	600 MHz (1200 MT/s)	600 MHz (1200 MT/s)
User Clock Frequency	150 MHz	150 MHz
PLL Reference Clock	50 MHz	50 MHz
Row Address Width	17 bits	16 bits
Column Address Width	10 bits	10 bits
Bank Address / Bank Group	2 / 2 bits	2 / 2 bits
ECC	Enabled	Enabled

DDR4 Timing Parameters Both memory interfaces share common timing parameters optimized for DDR4-2400 operation, as shown in Table 55.

Table 55: DDR4 Timing Parameters

Parameter	Value	Description
CAS Latency (CL)	12	Read latency in clock cycles
CAS Write Latency (CWL)	11	Write latency in clock cycles
tRCD	13.75 ns	Row to Column Delay
tRP	13.75 ns	Row Precharge Time
tRAS	32 ns	Row Active Time
tRC	45.75 ns	Row Cycle Time
tRFC	350 ns	Refresh Cycle Time
tREFI	7.8 μ s	Refresh Interval
tFAW	46.7 ns	Four Activate Window
tWR	15 ns	Write Recovery Time

7.4.2 DDR AXI4 Arbiter

Each DDR4 memory bank has a dedicated DDR AXI4 Arbiter instance [4] that provides the AXI4 interface between the fabric logic and the DDR4 MIG. Table 56 shows the configuration for both arbiters.

Table 56: DDR4 AXI4 Arbiter Configuration

Parameter	16 GB Arbiter	8 GB Arbiter
AXI Data Width	512 bits	256 bits
AXI Address Width	39 bits	38 bits
AXI ID Width	4 bits	4 bits
Read Channels	1	1
Write Channels	1	1

Each arbiter is configured with one read channel and one write channel. The write channel receives camera frame data from the DMA Write module, while the read channel serves DMA Read requests for data export via UDP or PCIe. The arbiter connects directly to the CoreDDR.LiteAXI wrapper of the DDR4 MIG, operating in the 150 MHz DDR4 user clock domain.

DDR Arbiter Timing Waveform The diagram 16 show the signaling interface Arbiter write channel and diagram 17 shows signaling interface for Arbiter's read channel. and read channel. For further information please refer to [4].

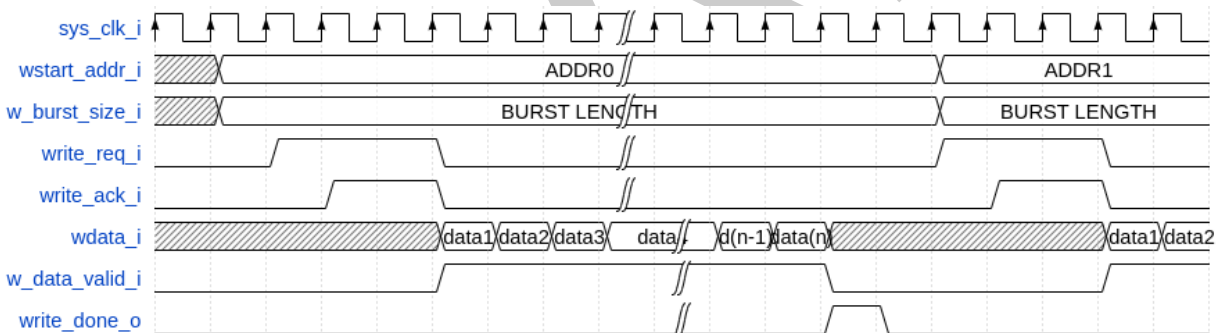


Figure 16: Timing Diagram for Writing into Memory [4]

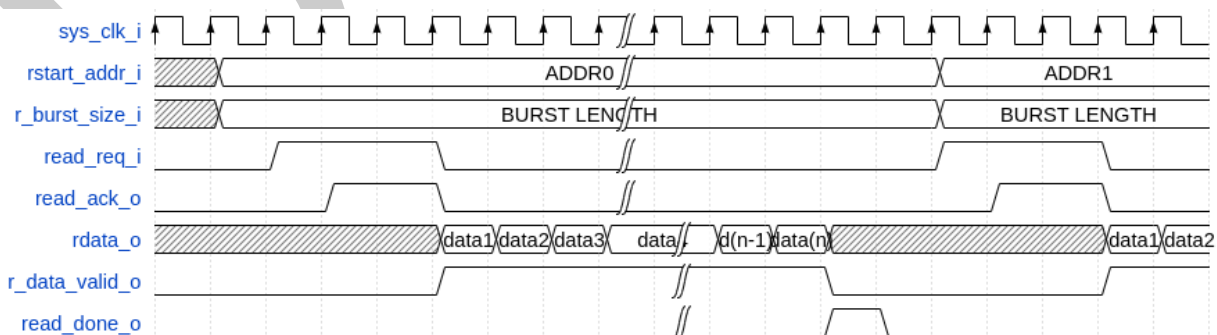


Figure 17: Timing Diagram for Reading from Memory [4]

7.4.3 DMA Write

The DMA Write module transfers camera frame data from the SLVS-EC receiver to DDR4 memory. It operates across two clock domains: the pixel clock domain (79.2 MHz) for receiving camera data and the DDR4 user clock domain (150 MHz) for memory writes.

Module Hierarchy The DMA Write IP consists of the following submodules:

- **dma_write_ddr4_16gb / dma_write_ddr4_8gb**: Top-level wrappers for each DDR4 bank
- **dma_write_ctrl**: Main controller instantiating the width converter and sub-controllers
- **wconv_4in_3out**: Width converter (384-bit to 512-bit)
- **dma_write_recv_ctrl**: Receive controller in pixel clock domain
- **dma_write_send_ctrl**: Send controller in DDR4 clock domain
- **COREFIFO**: Asynchronous FIFO for clock domain crossing

Width Conversion The camera interface outputs 384-bit data (32 pixels $\times$ 12 bits). The `wconv_4in_3out` module converts this to 512-bit words for DDR4 using a 4:3 ratio ($384 \times 4 = 512 \times 3$):

- Cycle 0: Buffer 384 bits, no output
- Cycle 1: Output 512 bits (128 new + 384 buffered), buffer 256 bits
- Cycle 2: Output 512 bits (256 new + 256 buffered), buffer 128 bits
- Cycle 3: Output 512 bits (384 new + 128 buffered), buffer empty

Receive Controller The `dma_write_recv_ctrl` operates in the pixel clock domain and counts the number of 512-bit chunks written to the FIFO per camera line. It detects the rising edge of `line_valid` to start counting and the falling edge to finalize the count. The chunk count is passed to the send controller via a handshake interface.

Send Controller The `dma_write_send_ctrl` operates in the DDR4 clock domain and manages AXI4 burst writes to the arbiter. Key features:

- Supports burst lengths up to 256 beats (AXI4 maximum)
- Splits large line transfers into multiple bursts if needed
- Maintains frame and line address tracking
- Configurable line gap (`h_size_byte`) for memory stride
- 10 ms watchdog timeout for error detection

The send controller FSM states:

1. **IDLE**: Wait for `frame_valid`
2. **WAIT_VALID_CHUNK**: Wait for chunk count from receive controller
3. **SET_BEAT_COUNTER**: Calculate burst length
4. **WRITE_TRIG**: Assert `arb_write_req` and wait for `arb_write_ack`
5. **WRITING**: Stream data from FIFO to arbiter
6. **CHECK_REMAINING**: Check if more bursts needed for current line
7. **SET_NEXT_LINE_ADDR**: Advance to next line address

Memory Address Mapping Frame data is organized in DDR4 memory using the following address structure:

- 16 GB DDR4: 9-bit frame index (512 frames), 25-bit frame address space (~32 MB per frame)
- 8 GB DDR4: 8-bit frame index (256 frames), 25-bit frame address space (~32 MB per frame)

7.4.4 DMA Write APB Register Interface

Table 57 shows the DMA Write APB registers.

Table 57: DMA Write APB Register Map

Offset	Name	Width	Access	Description
0x00	CLEAR_INDEX	1 bit	R/W	Reset frame index counter to 0
0x04	H.SIZE_BYTE	14 bits	R/W	Line stride in bytes (memory gap between lines)
0x08	FRAME_INDEX	8/9 bits	RO	Current frame index being written
0x0C	CORE_READY	1 bit	RO	DMA core ready for new frame
0x10	FRAME_WRITE_DONE	1 bit	RO	Pulse when frame write completes
0x14	TIMEOUT_ERR	32 bits	RO	Timeout error counter

7.4.5 DMA Read for UDP export

The DMA Read module transfers frame data from DDR4 memory to downstream consumers (UDP export). It operates across two clock domains: the DDR4 user clock domain (150 MHz) for memory reads and the UDP controller clock domain (100 MHz) for data output.

Module Hierarchy The DMA Read IP consists of the following submodules:

- **dma_read_ddr4_16gb / dma_read_ddr4_8gb**: Top-level wrappers for each DDR4 bank
- **dma_read_ctrl**: Main controller handling AXI4 read transactions and FIFO management
- **COREFIFO**: Asynchronous FIFO for clock domain crossing and width conversion (512-bit to 32-bit for 16 GB, 256-bit to 32-bit for 8 GB)

Controller Interface The DMA Read module receives read requests from an external controller (e.g., UDP export logic) via a handshake interface:

- **ctrl.info_valid**: Controller asserts to indicate valid address/burst parameters
- **ctrl.burst_count**: Number of AXI4 beats to read (9 bits, up to 512 beats)
- **ctrl.read_addr**: DDR4 start address for the read operation
- **dma_read_req**: Controller requests a DMA read transaction
- **dma_ready**: DMA signals it is ready to accept a new request
- **dma_read_ack**: DMA acknowledges the read request

Read Controller FSM The `dma_read_ctrl` operates in the DDR4 clock domain with the following states:

1. **IDLE**: Wait for `dma_read_req` rising edge, capture burst parameters
2. **READ_TRIG**: Assert `arb_read_req` and wait for `arb_read_ack`
3. **READING**: Stream data from arbiter to FIFO until `arb_read_done`
4. **WAIT_EXPORT_DONE**: Wait for FIFO to empty (data consumed by controller)
5. **CLEAR_STATE**: Reset state when clear signal is asserted

Data Flow Read data flows through an asynchronous FIFO that provides:

- Clock domain crossing from DDR4 clock (150 MHz) to UDP controller clock (100 MHz)
- Width conversion from DDR4 data width (512/256 bits) to 32-bit AXI-Stream output
- Buffering to decouple DDR4 burst reads from downstream consumption rate

The output uses an AXI-Stream interface (`m_axis_dma_tvalid`, `m_axis_dma_tready`, `m_axis_dma_tdata`) for flow-controlled data transfer to the export logic.

FIFO Management The FIFO can be cleared via two mechanisms:

- **dma_fifo_clear**: Per-transaction FIFO clear from the controller
- **clear**: Global clear from APB register (also resets timeout error counter)

Both clear signals are synchronized across clock domains before asserting the FIFO reset.

Timeout Protection The controller implements a 10 ms watchdog timeout during READ_TRIG, READING, and WAIT_EXPORT_DONE states. If a timeout occurs, the FSM returns to IDLE and increments the `timeout_err` counter. This prevents the DMA from stalling indefinitely due to arbiter or downstream issues.

7.4.6 DMA Read APB Register Interface

Table 58 shows the DMA Read APB registers.

Table 58: DMA Read APB Register Map

Offset	Name	Width	Access	Description
0x00	CLEAR	1 bit	R/W	Reset DMA Read controller and clear timeout error counter
0x04	DMA_TIMEOUT_ERR	32 bits	RO	Timeout error counter

7.5 UDP Export Data Architecture

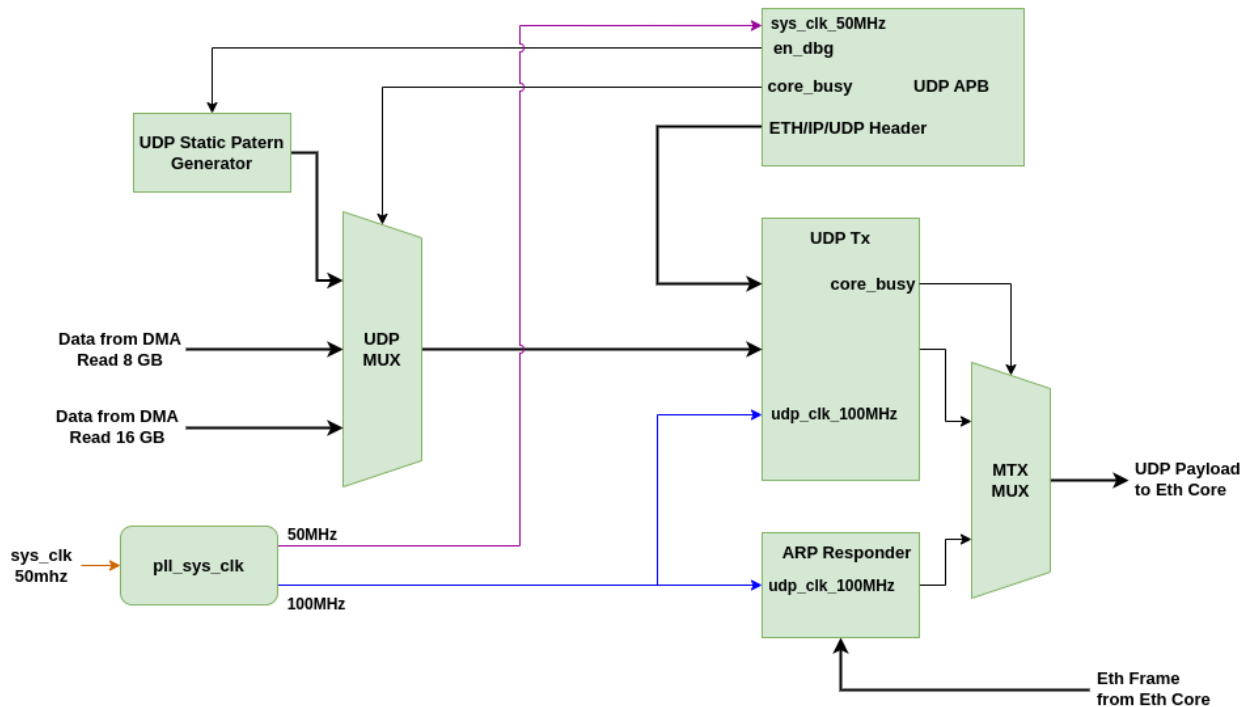


Figure 18: Farsight Avionics UDP Data Out Architecture

The UDP Export subsystem transfers frame data from DDR4 memory to the Ethernet MAC for transmission over UDP. It consists of two main components: the UDP DMA Controller (`udp_dmactrl_ip`) and the UDP TX IP (`udp_ip`). Both operate at 100 MHz in the UDP controller clock domain.

7.5.1 UDP DMA Controller

The UDP DMA Controller orchestrates frame data transfer from DDR4 memory to the UDP TX core. It consists of two submodules:

- **frame_xfer_ctrl**: Frame-level controller that requests entire image frames line-by-line
- **udp_dmactrl**: Packet-level controller that segments lines into UDP packets

Frame Transfer Controller The `frame_xfer_ctrl` module manages the transfer of complete image frames from DDR4 memory. It receives frame parameters from software via APB registers and issues DMA read requests line-by-line.

Key parameters:

- **frame_index**: DDR4 frame buffer index to read from
- **h_size_beat**: Number of AXI4 beats per line (max 511)
- **h_size_byte**: Line stride in bytes for address calculation
- **v_size_line**: Number of lines per frame (max 8192)

The controller FSM states:

1. **IDLE**: Wait for `frame_read_req` from software
2. **CLEAR_DMA_FIFO**: Reset DMA FIFO before starting transfer

3. **WAIT_FOR_DMA_READY**: Wait for DMA Read module to be ready
4. **SET_BEAT_COUNTER**: Calculate burst size (splits lines >256 beats)
5. **READ_LINE_TRIG**: Request line read from UDP DMA controller
6. **WAIT_LINE_XFER_DONE**: Wait for line transfer completion
7. **CHECK_H_CHUNK**: Check if more horizontal chunks needed
8. **CHECK_V_COUNT**: Advance to next line or complete frame

UDP DMA Controller The `udp_dmactrl` module segments line data into UDP packets and interfaces with the UDP TX core. It supports both standard (1500 byte) and jumbo (4000 byte) Ethernet frames.

Packet size limits:

- Standard frame: $363 \text{ words} \times 4 \text{ bytes} + 6 \text{ bytes header} = 1458 \text{ bytes payload (1500 byte total frame)}$
- Jumbo frame: $988 \text{ words} \times 4 \text{ bytes} + 6 \text{ bytes header} = 3958 \text{ bytes payload (4000 byte total frame)}$

Each UDP packet includes a 6-byte header:

- 2 bytes: Checksum placeholder (0x0000)
- 2 bytes: Packet start token ("PK" = 0x504B)
- 2 bytes: Line index within the frame
- 2 bytes: Packet sequence count within the line

The controller FSM states:

1. **IDLE**: Wait for line read request from frame controller
2. **CALC_UDP_PYL_SIZE**: Calculate packet payload size
3. **WAIT_FOR_DMA_ACK**: Wait for DMA Read acknowledgment
4. **SEND_SOF_REQ**: Request start of frame from UDP TX
5. **WAIT_FOR_PYL_ACPT**: Wait for UDP TX to accept payload
6. **SEND_FIRST**: Send checksum placeholder and start token
7. **SEND_SECOND**: Send line index and packet count
8. **SEND_PAYLOAD**: Stream DMA data to UDP TX
9. **SEND_LAST**: Send final word with `tlast` assertion
10. **WAIT_FOR_ACK**: Wait for UDP TX end-of-frame acknowledgment
11. **CHECK_FOR_CHUNK**: Check if more packets needed for current line

7.5.2 UDP DMA Controller APB Register Interface

Table 59 shows the UDP DMA Controller APB registers.

Table 59: UDP DMA Controller APB Register Map

Offset	Name	Width	Access	Description
0x00	CLEAR	1 bit	R/W	Reset controller and clear errors
0x04	FRAME_INDEX	8 bits	R/W	Frame buffer index to read
0x08	FRAME_READ_DONE_COUNT	32 bits	RO	Number of frames transferred
0x0C	FRAME_READ_REQ	1 bit	R/W	Trigger frame read (rising edge)
0x10	H_SIZE_BEAT	9 bits	R/W	AXI4 beats per line
0x14	H_SIZE_BYTE	14 bits	R/W	Line stride in bytes
0x18	JUMBO_EN	1 bit	R/W	Enable jumbo frames (4000 bytes)
0x1C	V_SIZE_LINE	13 bits	R/W	Number of lines per frame
0x20	DMA_TIMEOUT_ERR	32 bits	RO	DMA timeout error counter
0x24	PYL_ACPT_ERR	32 bits	RO	Payload accept timeout counter
0x28	SEND_LAST_ERR	32 bits	RO	Send last timeout counter
0x2C	SEND_PYL_ERR	32 bits	RO	Send payload timeout counter
0x30	SOF_REQ_ERR	32 bits	RO	SOF request timeout counter
0x34	WAIT_ACK_ERR	32 bits	RO	EOF ack timeout counter
0x38	FRAME_XFER_TIMEOUT_ERR	32 bits	RO	Frame transfer timeout counter
0x3C	RESET_DONE_COUNTER	1 bit	R/W	Reset frame done counter

7.5.3 UDP TX IP

The UDP TX IP constructs complete Ethernet frames with IP and UDP headers, then transmits them to the PolarFire CoreTSE MAC. It accepts payload data via an AXI-Stream interface and network parameters via APB registers.

Module Hierarchy The UDP TX IP consists of:

- **udp_tx_top**: Top-level wrapper containing the TX core and backpressure FIFO
- **udp_tx**: Main controller that constructs Ethernet/IP/UDP headers and manages MAC interface
- **udp_mux**: Multiplexer to select between DDR4 8 GB, DDR4 16 GB, or debug data sources
- **COREFIFO_MAC_BACKPRES**: FIFO for MAC backpressure handling

Frame Construction The UDP TX core constructs a complete Ethernet frame with the following structure:

- **Ethernet Header** (14 bytes): Destination MAC, Source MAC, EtherType (0x0800 for IPv4)
- **IP Header** (20 bytes): Version 4, IHL 5, TTL 64, Protocol 17 (UDP), IP checksum calculated in hardware
- **UDP Header** (8 bytes): Source port, Destination port, Length, Checksum (set to 0)
- **Payload**: Data from UDP DMA Controller

The IP header checksum is calculated over two clock cycles using one's complement sum of all 16-bit header words.

MAC Interface The UDP TX interfaces with the PolarFire CoreTSE MAC using the following signals:

- **MTXSOF**: Start of frame indicator
- **MTXRDY**: Data ready (valid)
- **MTXDAT**: 32-bit data output (byte-swapped for network byte order)
- **MTXEOF**: End of frame indicator
- **MTXBYTEVALID**: Valid bytes in last word
- **MTXACPT**: MAC ready to accept data (backpressure signal)

Backpressure Handling The CoreTSE MAC may temporarily stop accepting data. The UDP TX handles this via:

- A FIFO buffer between payload input and MAC output
- Backpressure state tracking to hold data when MAC deasserts MTXACPT
- Automatic retry when MAC resumes accepting data

Frame Gap A configurable frame gap (default 1000 clock cycles at 100 MHz = 10 μ s) is enforced between consecutive frames. The CoreTSE MAC requires a minimum gap of approximately 20 μ s for reliable operation.

Timeout Protection The UDP TX implements a 10-second watchdog timeout (configurable via MAX_TIMEOUT_USEC). If any state exceeds the timeout, the core sends an error marker (0xEFBADBAD) with EOF and returns to IDLE.

UDP TX FSM States

1. **IDLE**: Wait for `sof_req` from DMA controller
2. **PREPARE**: Capture header values from registers
3. **CHECKSUM.CALC.P1**: Sum IP header words
4. **CHECKSUM.CALC.P2**: Compute one's complement checksum
5. **SEND.HEADER**: Transmit 42-byte header (10 words + 2 bytes)
6. **SEND.PAYLOAD**: Stream payload data to MAC
7. **FLUSH.REM.BUFFER**: Empty remaining FIFO data
8. **SEND.PAYLOAD.LAST**: Send final word with EOF
9. **FRAME.GAP.WAIT**: Wait for inter-frame gap
10. **WD.TIMEOUT.ERR**: Error recovery state

7.5.4 UDP MUX

The `udp_mux` module selects the data source for UDP transmission. Table 60 shows the available selections.

Table 60: UDP MUX Selection Values

Value	Name	Description
0b00	DISABLED	UDP TX disabled, all outputs zero
0b01	DBG_EN	Debug static UDP mode for testing
0b10	UDP_FOR_DDR4_8GB	Route DDR4 8 GB data to UDP TX
0b11	UDP_FOR_DDR4_16GB	Route DDR4 16 GB data to UDP TX

7.5.5 UDP TX APB Register Interface

Table 61 shows the UDP TX APB registers for network configuration.

Table 61: UDP TX APB Register Map

Offset	Name	Width	Access	Description
0x00	UDP_DST_PORT	16 bits	R/W	UDP destination port
0x04	UDP_SRC_PORT	16 bits	R/W	UDP source port
0x08	DST_IP	32 bits	R/W	Destination IPv4 address
0x0C	SRC_IP	32 bits	R/W	Source IPv4 address
0x10	DST_MAC_MSB	16 bits	R/W	Destination MAC [47:32]
0x14	DST_MAC_LSB	32 bits	R/W	Destination MAC [31:0]
0x18	SRC_MAC_MSB	16 bits	R/W	Source MAC [47:32]
0x1C	SRC_MAC_LSB	32 bits	R/W	Source MAC [31:0]
0x20	WD_TIMEOUT_ERR_COUNT	32 bits	RO	Watchdog timeout counter
0x24	MUX_SEL	2 bits	R/W	UDP MUX source select
0x28	UDP_CLR	1 bit	R/W	Clear UDP TX core
0x2C	DBG_EN_JUMBO	1 bit	R/W	Enable jumbo frames in debug mode
0x30	DBG_MAX_DELAY	32 bits	R/W	Debug mode delay setting
0x40	FRAME_GAP	32 bits	R/W	Inter-frame gap in clock cycles

7.6 Ethernet MAC

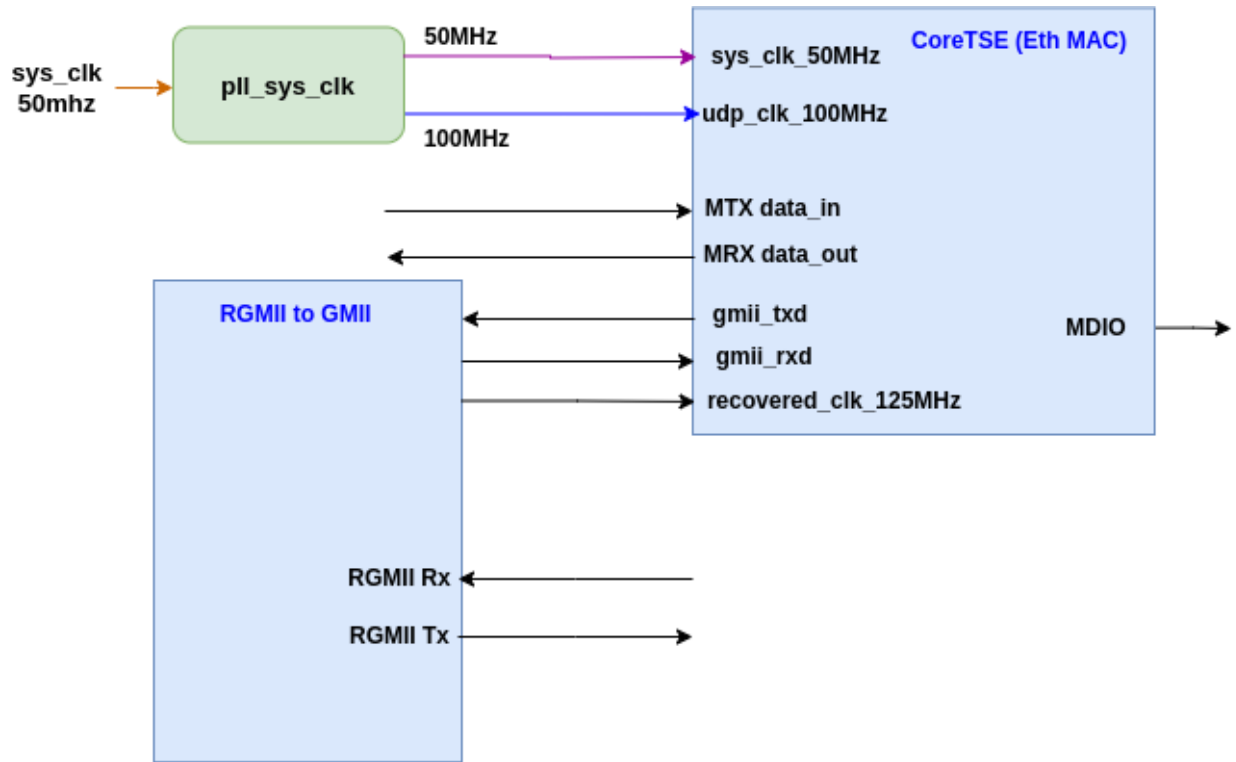


Figure 19: Farsight Avionics Ethernet MAC Architecture

The Farsight Avionics board includes two Ethernet interfaces (ETH1 and ETH2) based on Microchip VSC8541XMV-05 Gigabit Ethernet PHYs [2]. The FPGA provides MAC layer functionality via the PolarFire CoreTSE IP [3], with RGMII-to-GMII conversion for PHY interfacing.

7.6.1 Ethernet Hierarchy Components

Each Ethernet interface consists of the following components:

- **CoreTSE MAC:** Microchip CoreTSE v4.0.124 Triple-Speed Ethernet MAC IP
- **PF_RGMII_TO_GMII:** PolarFire RGMII to GMII bridge IP (v1.3.109)
- **CoreGPIO (Control):** GPIO output for PHY control signals
- **CoreGPIO (Status):** GPIO input for PHY status signals
- **BIBUF:** Bidirectional buffer for MDIO interface

7.6.2 CoreTSE MAC Configuration

Table 62 shows the CoreTSE MAC configuration parameters.

7.6.3 RGMII to GMII Bridge

The PF_RGMII_TO_GMII IP converts between the 4-bit RGMII interface (to the external PHY) and the 8-bit GMII interface (to the CoreTSE MAC). The bridge is configured with:

- Clock-to-data alignment: Aligned

Table 62: CoreTSE ETH1 MAC Configuration

Parameter	Value
Interface Mode	GMII (not TBI)
MDIO PHY Address	18
Max Packet Size	8192 bytes (jumbo frame support)
Statistics Counters	Enabled
Wake-on-LAN	Disabled
SLIP Mode	Disabled

- RX clock source: Global clock network

7.6.4 Ethernet MAC Clock Domains

The Ethernet subsystem operates across two clock domains:

- **APB Clock (50 MHz):** Used for register access to CoreTSE MAC, control GPIO, and status GPIO
- **MAC Data Clock (100 MHz):** Used for the MAC TX/RX data interfaces (MTXCLK, MRXCLK)

7.6.5 PHY Control GPIO Interface

The Ethernet PHY is controlled via a CoreGPIO instance accessible from the RISC-V firmware. Table 63 shows the control GPIO output bits.

Table 63: Ethernet PHY Control GPIO (Output)

Bit	Name	Default	Description
0	eth1_phy_rst_n	0	PHY active-low reset
1	eth1_ctrl_comma_mode	0	PHY comma mode control
2	eth1_ctrl_clk_squelch_in	0	PHY clock squelch input

7.6.6 PHY Status GPIO Interface

PHY status signals are read via a separate CoreGPIO instance. Table 64 shows the status GPIO input bits.

Table 64: Ethernet PHY Status GPIO (Input)

Bit	Name	Description
0	eth1_stat_fastlink_fail	Fast link failure indicator
1	eth1_stat_mdint	Management data interrupt
2	eth1_stat_rcvrd_clk	Recovered clock status
3	eth1_stat_clkout	PHY clock output status

7.6.7 Ethernet MAC Data Interface

The CoreTSE MAC provides separate TX and RX data interfaces for fabric connectivity:

Transmit Interface (from UDP TX IP)

- **MTXSOF:** Start of frame indicator
- **MTXRDY:** Data valid signal

- **MTXDAT**: 32-bit transmit data
- **MTXEOF**: End of frame indicator
- **MTXBYTEVALID**: Valid bytes in last word (2 bits)
- **MTXACPT**: MAC ready to accept data (backpressure)

Receive Interface

- **MRXSOF**: Start of frame indicator
- **MRXRDY**: Data valid signal
- **MRXDAT**: 32-bit receive data
- **MRXEOF**: End of frame indicator
- **MRXBYTEVALID**: Valid bytes in last word (2 bits)
- **MRXACPT**: Fabric ready to accept data
- The ETH1 MAC is dedicated to image data transfer and is controlled exclusively by the FPGA gateway (UDP TX IP). It is not directly accessible from the RISC-V firmware.
- The MAC supports Gigabit Ethernet (1000 Mbps), however due to MAC FIFO handshaking constraints in the current design, the effective data rate is limited to approximately 800 Mbps.
- The PHY control and status GPIO interfaces are accessible from the RISC-V firmware for PHY initialization and monitoring.

7.7 Housekeeper

Farsight Avionics Housekeeper consists of two FPGA chips for managing and monitoring power status of the system. The ProASIC3 FPGA and PolarFire FPGA.

7.7.1 ProASIC3 FPGA Housekeeper Architecture

The ProASIC3 FPGA Housekeeper is responsible for controlling and monitoring power to various subsystems on the Farsight Avionics board. It manages 33 power sources organized into 11 power regions, each with dedicated state machines for sequencing and fault detection.

Power Source Interface Each power source on the board provides the following signals:

- **Enable** (output, active high): Controlled by the FPGA to turn the power source on or off.
- **PGOOD** (input, active high): Indicates the power source has stabilized and is operating correctly.
- **NFAULT** (input, active low): Present on some sources to indicate a fault condition.

All PGOOD and NFAULT inputs pass through a 5 μ s glitch filter to reject transient noise before being processed by the housekeeper logic.

Table 65: PA3 Housekeeper Region Information

Boot	Power Region	Power Source	Fault?	Critical	SW Ctrl	Mode When First Enabled	Nominal Boot Time
1	Step Down Conv.	2V2	Yes	Yes	No	—	2.6 ms
2	Step Down Conv.	3V0	Yes	Yes	No	—	2.0 ms
3	Step Down Conv.	4V0	Yes	Yes	No	—	2.0 ms
4	DDR8	2V5	Yes	No	No	—	0.704 ms
5	DDR8	1V2	Yes	No	No	—	0.52 ms
6	DDR8	0V6	No	No	No	—	0.40 ms
7	DDR16	2V5	Yes	No	No	—	1.23 ms
8	DDR16	1V2	Yes	No	No	—	0.91 ms
9	DDR16	0V6	No	No	No	—	0.40 ms
10	FPGA	1V0	Yes	Yes	No	BOOT.MODE	0.64 ms
11	FPGA	1V0A	No	Yes	No	BOOT.MODE	2.3 ms
12	FPGA	1V25A	No	Yes	No	BOOT.MODE	2.3 ms
13	FPGA	1V8	No	Yes	No	BOOT.MODE	2.3 ms
14	FPGA	1V8 IMX	No	Yes	No	BOOT.MODE	2.3 ms
15	FPGA	2V5A	No	Yes	No	BOOT.MODE	2.3 ms
16	FPGA	3V3 B4	No	Yes	No	BOOT.MODE	2.3 ms
17	FPGA	3V3 B5	No	Yes	No	BOOT.MODE	2.3 ms
1	LVDS	3V3	No	No	Yes	—	2.3 ms
1	Eth1	1V0	No	No	Yes	—	2.3 ms
2	Eth1	1V0A	No	No	Yes	—	2.3 ms
3	Eth1	2V5A	No	No	Yes	—	2.3 ms
4	Eth1	3V3	No	No	Yes	—	2.3 ms
1	Eth2	1V0	No	No	Yes	—	2.3 ms
2	Eth2	1V0A	No	No	Yes	—	2.3 ms
3	Eth2	2V5A	No	No	Yes	—	2.3 ms
4	Eth2	3V3	No	No	Yes	—	2.3 ms
1	Stepper Primary Motor	28V0	Yes	No	Yes	Only when in use	N/A
1	Stepper Secondary Motor	28V0	Yes	No	Yes	Only when in use	N/A
1	LVDT	15V0	No	No	Yes	—	2.3 ms
1	IMX	1V1	Yes	No	Yes	IDLE.MODE	1.22 ms
2	IMX	1V8	No	No	Yes	IDLE.MODE	2.3 ms
3	IMX	2V9	No	No	Yes	IDLE.MODE	0.45 ms
4	IMX	3V3	No	No	Yes	IDLE.MODE	5.1 ms

Power Region Types Power regions are classified into two categories:

- **Hardware-Controlled (Critical) Regions:** Four regions that boot automatically at power-up in a fixed sequence. These include the step-down converters, PolarFire FPGA core rails, and DDR memory. A failure in any critical region triggers a controlled system shutdown.

- **Software-Controlled Regions:** Regions controlled by the PolarFire firmware after the system has successfully booted. These include the camera sensor, Ethernet PHYs, stepper motors, and LVDT interface. Each region can be independently enabled or disabled via software commands.

Boot Sequence The power-up sequence proceeds as follows:

1. Hardware-controlled regions boot sequentially in a predefined order, with each region waiting for the previous region's PGOOD before starting.
2. If a power source fails to assert PGOOD within twice the expected boot time, the region is disabled for 200 ms and re-attempted.
3. Each region is allowed up to four boot attempts before being declared inoperable.
4. Once all critical regions are operational, software-controlled regions can be enabled by the PolarFire firmware.

Latchup Detection and Handling A latchup condition is detected when:

- A PGOOD signal goes low after the power source was successfully booted, or
- An NFAULT signal goes low after the source was enabled.

The response to a latchup depends on the region type:

- **Hardware-controlled region latchup:** Causes immediate and permanent shutdown of all power sources until the next power cycle.
- **Software-controlled region latchup:** Causes immediate shutdown of only that specific power region. The region can be re-enabled by software after the fault is cleared.

Controlled Shutdown A controlled shutdown is initiated when:

- The EPS eFuse PGOOD signal has been low for more than 5 μ s, or
- A critical power region fails to boot after all retry attempts.

During controlled shutdown:

1. All software-controlled regions are disabled simultaneously.
2. Hardware-controlled regions are then disabled one by one in reverse boot order.
3. Each power source within a region is disabled sequentially, waiting for PGOOD to fall before proceeding to the next source.

Failure Reporting During normal operation, the ProASIC3 operates in pass-through mode, routing UART traffic directly between the PolarFire and the payload bus. If a critical failure occurs and the PolarFire is powered down, the ProASIC3 takes control of the UART and broadcasts a failure status message approximately once per second. The message format is shown in Table 66.

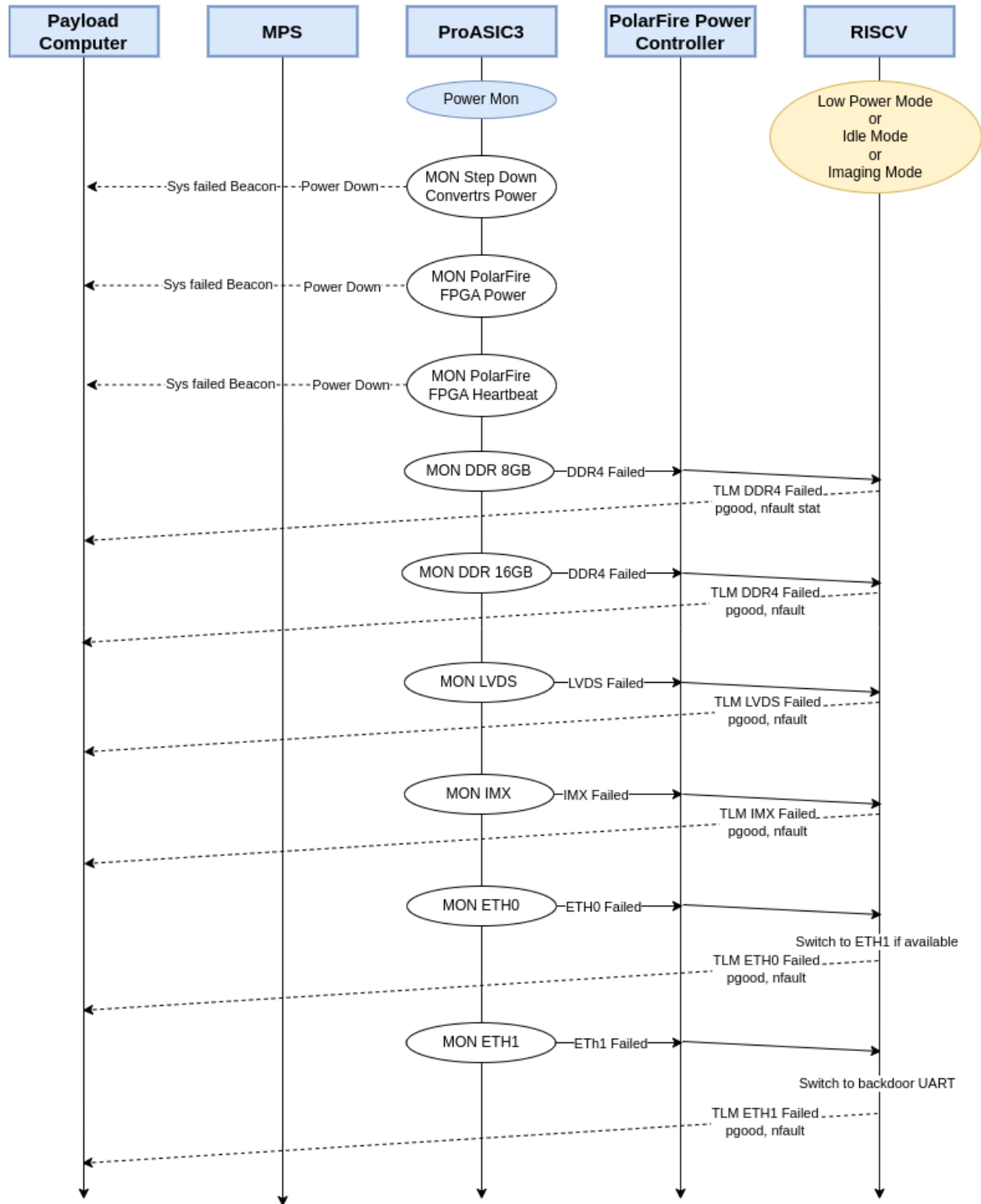


Figure 20: Farsight Hardware Powerup Sequence Diagram

Table 66: Housekeeper UART Failure Message Format

Byte	Region	b7	b6	b5	b4	b3	b2	b1	b0
0	Header	0xDE							
1	Header	0xAD							
2	Header	0xBE							
3	Header	0xEF							
4	FPGA pgood	3V3_B5	3V3_B4	2V5A	1V8_IMX	1V8	1V25A	1V0A	1V0
5	FPGA / LVDS	Reserved							
6	LVDS	Reserved							
7	DDR16	Reserved		0V6_pgood	1V2_pgood	2V5_pgood	1V2_nfault	2V5_nfault	Reserved
8	DDR8	Reserved		0V6_pgood	1V2_pgood	2V5_pgood	1V2_nfault	2V5_nfault	Reserved
9	Step Down	Reserved	4V0_pgood	3V0_pgood	2V2_pgood	4V0_nfault	3V0_nfault	2V2_nfault	Reserved
10	Trailer	0xBA							
11	Trailer	0xDD							
12	Trailer	0xFE							
13	Trailer	0xED							

7.7.2 PolarFire FPGA Housekeeper (HK) Architecture

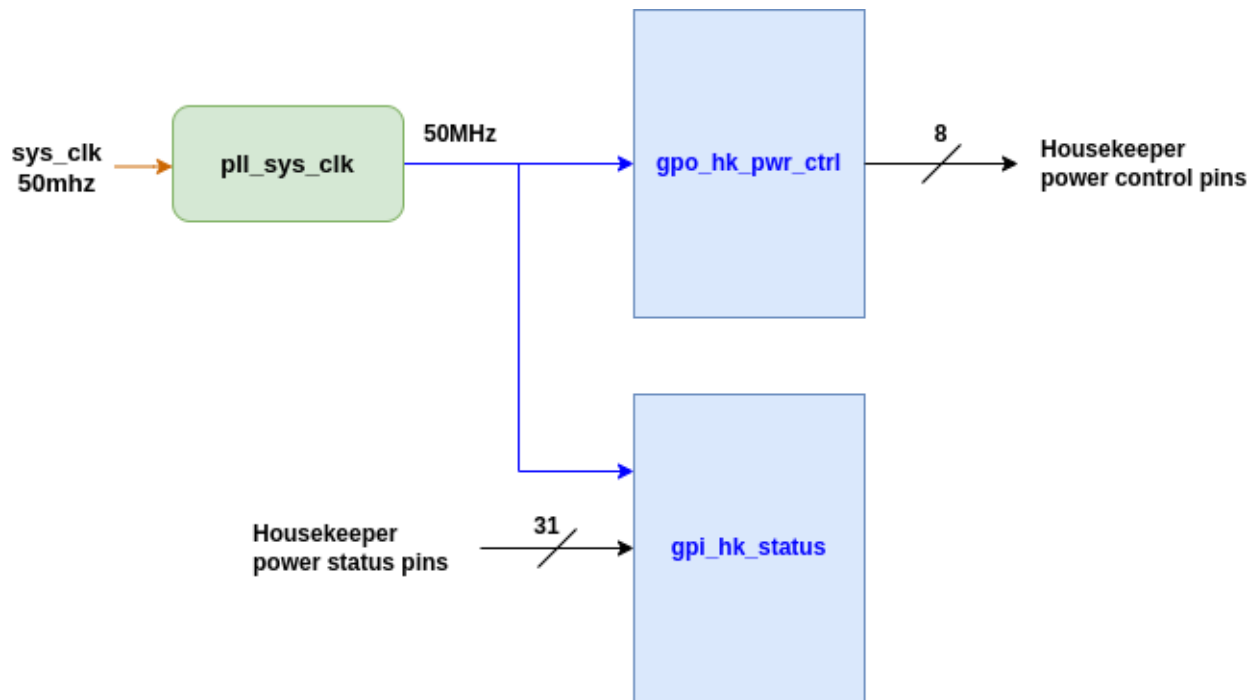


Figure 21: Farsight Avionics PolarFire FPGA Housekeeper Component

The PolarFire FPGA Housekeeper provides the interface between the RISC-V firmware and the system power management. Figure 21 shows the overall structure of the HK. Its primary functions are:

- Receive power status signals from the ProASIC3 FPGA Housekeeper
- Control power enable signals for software-controlled power domains
- Provide miscellaneous inter-FPGA communication signals

The housekeeper hierarchy consists of two CoreGPIO instances accessible via APB:

- **gpi_hk_status**: 32-bit input GPIO for power status monitoring
- **gpo_hk_pwr_ctrl**: 32-bit output GPIO for power enable control

Power Status GPIO (Input) Table 67 shows the power status input bits received from the ProASIC3 Housekeeper and other system status signals.

Power Control GPIO (Output) Table 68 shows the power enable control bits that are sent to the ProASIC3 Housekeeper to control power domain enables.

Camera Power Interlocks The camera subsystem includes automatic interlocks based on the cam_pwr_status signal:

- **cam_spi_tribuff_en**: Directly driven by cam_pwr_status. The camera SPI signals are tri-stated when camera power is off.
- **cam_buff_en.n**: Inverted cam_pwr_status. The camera buffer is enabled (active low) only when camera power is good.

Table 67: Housekeeper Power Status GPIO (Input)

Bit	Name	Description
0	pa3_pwr_status	ProASIC3 power good status
1	step_down_pwr_status	Step-down regulator power good
2	ddr8gb_pwr_status	DDR4 8 GB power good
3	ddr16gb_pwr_status	DDR4 16 GB power good
4	pf_pwr_status	PolarFire power good
5	lvds_pwr_status	LVDS interface power good
6	eth1_pwr_status	Ethernet 1 PHY power good
7	eth2_pwr_status	Ethernet 2 PHY power good
8	stepper_pri_pwr_status	Primary stepper motor power good
9	stepper_sec_pwr_status	Secondary stepper motor power good
10	lvdt_pwr_status	LVDT sensor power good
11	cam_pwr_status	Camera power good (also controls SPI tri-state buffer)
12–27	pa3_to_pf_misc[0:15]	Miscellaneous ProASIC3 to PolarFire signals
28	Reserved	Tied to GND
29–31	pa3_fw_version[2:0]	ProASIC3 firmware version (3 bits)

Table 68: Housekeeper Power Control GPIO (Output)

Bit	Name	Default	Description
0	lvds_pwr_en	0	LVDS interface power enable
1	eth1_pwr_en	0	Ethernet 1 PHY power enable
2	eth2_pwr_en	0	Ethernet 2 PHY power enable
3	stepper_pri_pwr_en	0	Primary stepper motor power enable
4	stepper_sec_pwr_en	0	Secondary stepper motor power enable
5	lvdt_pwr_en	0	LVDT sensor power enable
6	cam_pwr_en	0	Camera power enable
7	cam_osc_en	0	Camera oscillator enable
8–31	Reserved	0	Unused

This ensures that camera communication interfaces are only active when the camera power domain is properly powered.

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38	Register: ETH1_SRC_MAC_ADDR_LOW_SYS_REG	28
39	Register: ETH1_DEST_MAC_ADDR_HIGH_SYS_REG	29
40	Register: ETH1_DEST_MAC_ADDR_LOW_SYS_REG	29
41	Register: ETH1_SRC_IP_ADDR_SYS_REG	29
42	Register: ETH1_DEST_IP_ADDR_SYS_REG	29
43	Register: ETH1_SRC_UDP_PORT_SYS_REG	30
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8.3 Acronyms

Acronym	Full name
FC	Flight Computer
PC	Payload Computer
FAV	Farsight Payload Avionic Computer
CSR	Control and Status Registers
TMTC	Telemetry and Telecommand
C&DH	Command and Data Handling
CMOS	Complementary Metal-Oxide-Semiconductor
ConOps	Concept of Operations
PCIESS	embedded PCIe controller subsystem
HAL	FAV Hardware Abstraction Layer Middleware
ECO	Engineering Change Order
FPGA	Field Programmable Gate Array
GEO	Geostationary Orbit
GPI	General Purpose Input
GPO	General Purpose Output
ICD	Interface Control Document
LEO	Low Earth Orbit
PWR	Power
SERDES	Serializer/De-serializer
OB	Optical Black
FD	Full Digital
CDS	Correlated Double Sampling
PGA	Programmable Gain Amplifier
EC	Embedded Clock
SLVS	Scalable Low-Voltage Signaling
SLVS-EC	Scalable Low-Voltage Signaling with Embedded Clock
GMRWT	Global Master Read Wait Time
GMTWT	Global Master Transfer Wait Time
NVM	None Volatile Memory
SHS	Shutter sweep time
RoHS	Restriction of Hazardous Substances
DEDOO	Dedicated Input/Output

Table 69: Farsight Payload Acronyms

MetaITEM NUMBER: CM-01979				
REV:	DESCRIPTION:	ECO:	DATE:	APPROVED:
0.1	Sent to customer for review		09/09/2024	J. Parker
0.2	Design rework sent to customer for review		12/13/2024	J. Parker
0.3	Register and command updates		06/01/2025	J. Parker
0.4	Boot mode, fdir details, register updates and command updates		06/17/2025	J. Parker
0.5	Minor register updates including addition of ip and mac registers for configuring the second ethernet as well as a proasic3 firmware version register		7/15/2025	J. Parker
0.6	Simplified tmtc and removal of events for initial release		10/16/2025	J. Parker
0.7	Changed the icd to match the actual implementation of the first release of the fpga and firmware		12/8/2025	S. Janamian
0.8	Added updated register list based on fpga and firmware implementation		02/7/2026	S. Janamian

8.4 References

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